IGCSE Physics 0625

2025 Compact Workbook · 10 multiple choice + 3 theory questions per topic

Name: _____

Class: _____

Cambridge IGCSE Physics 0625 — 2025 (Feb/Mar, May/Jun, Oct/Nov)

练习册

Contents

Topic · Subtopic · page

1 MOTION, FORCES AND ENERGY

1.1 Physical quantities and measurement techniques
1.2 Motion
1.3 Mass and weight
1.4 Density
1.5 Forces
1.6 Momentum
1.7 Energy, work and power
1.8 Pressure

2 THERMAL PHYSICS

2.1 Kinetic particle model of matter
2.2 Thermal properties and temperature
2.3 Transfer of thermal energy

3 WAVES

3.1 General properties of waves
3.2 Light
3.3 Electromagnetic spectrum
3.4 Sound

4 ELECTRICITY AND MAGNETISM

4.1 Simple phenomena of magnetism
4.2 Electrical quantities
4.3 Electric circuits
4.4 Electrical safety
4.5 Electromagnetic effects

5 NUCLEAR PHYSICS

5.1 The nuclear model of the atom
5.2 Radioactivity

6 SPACE PHYSICS

6.1 The Earth and the Solar System
6.2 Stars and the Universe

1 MOTION, FORCES AND ENERGY

13 questions · Paper 2 & Paper 4

1 A student determines the diameter of a metal ball.

Which apparatus is needed to get an accurate measurement?

A a ruler and a protractor

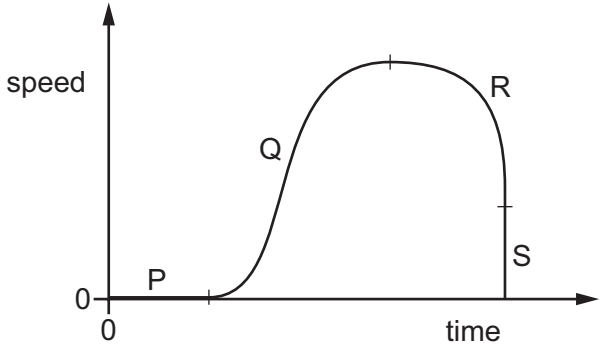
B a ruler only

C a protractor and a set square

D a ruler and two rectangular blocks

Date: _____ Score: _____

1 The speed–time graph for a racing car is divided into four sections, P, Q, R and S. The car starts the race but soon crashes into a wall of tyres.



Which sections of the graph show that there is an acceleration that is changing?

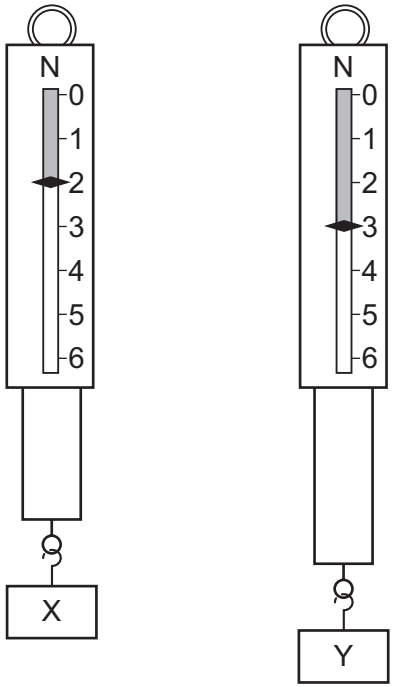
A P, Q, R and S

B Q, R and S only

C Q and R only

D R and S only

3 The diagrams show two metal blocks, X and Y, suspended from identical force meters. X and Y have identical dimensions.



Which statement about X and Y is correct?

A They have different volumes and different weights.

B They have different volumes and equal weights.

C They have equal volumes and equal weights.

D They have equal volumes and different weights.

4 A girl has a gold necklace.

Which apparatus does she need to find the density of the gold?

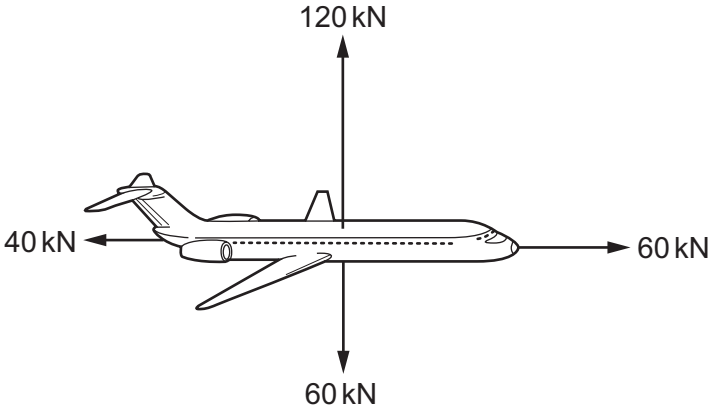
	balance	measuring cylinder	thermometer	ruler
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✓	✓	✗	✗
D	✓	✗	✗	✗

key

✓ = needed

✗ = **not** needed

2 The diagram shows the four forces acting on an aircraft in flight.



Which arrow shows the direction of the resultant of the four forces?

A



B



C



D



7 Which statement is **not** correct?

A the momentum of a system is conserved in all interactions

B impulse = change of momentum

C force acting = change of momentum × time

D momentum = mass × velocity

8 A machine is very efficient.

What does this mean?

A It produces a large amount of power.

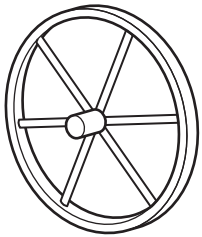
B It uses very little energy.

C It wastes very little energy.

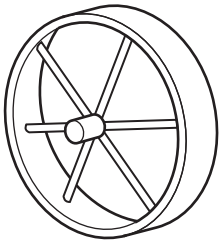
D It works very quickly.

9 A farmer has two carts. The carts have the same weight, but one has four narrow wheels and the other has four wide wheels.

narrow wheel



wide wheel



Which cart wheels sink less into soft ground and what is the reason?

	cart wheels	reason
A	narrow	greater pressure on the ground
B	narrow	less pressure on the ground
C	wide	greater pressure on the ground
D	wide	less pressure on the ground

Date: _____ Score: _____

1 Which instrument is most suitable to determine the volume of a small irregularly shaped stone?

A 30 cm ruler

B digital timer

C measuring cylinder

D tape measure

2 An athlete runs at a speed of 8 m / s for 10 s, and then at a speed of 6 m / s for 12 s.

Which calculation gives the average speed of the athlete in m / s?

A $\frac{8+6}{2}$

B $\frac{(8\times10)+(6\times12)}{22}$

C $\frac{(8\div10)+(6\div12)}{22}$

D $\frac{(10\div8)+(12\div6)}{22}$

1 **(a)** Circle the vector quantities in the list.

acceleration **mass** **speed** **time** **velocity**

[1]

(b) Fig. 1.1 shows the speed–time graph for a train travelling from station A to station B.

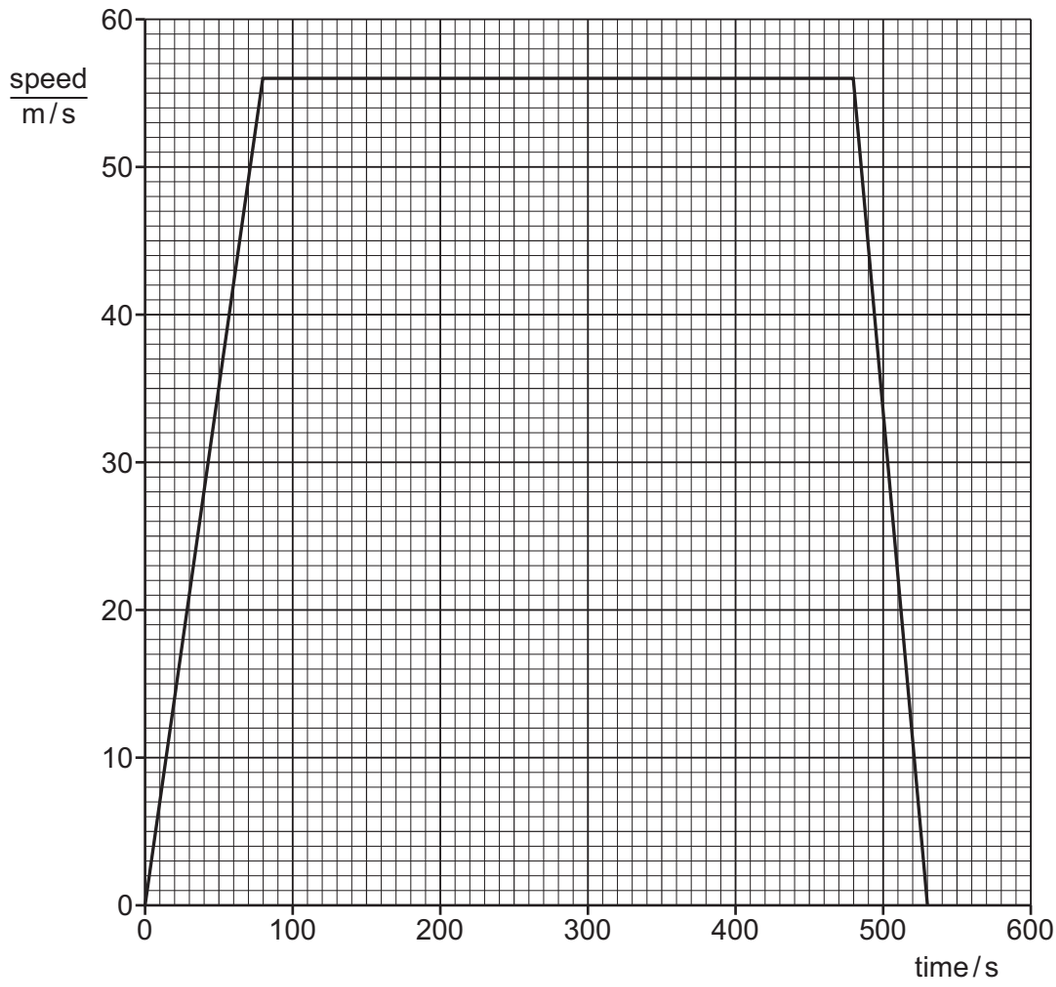


Fig. 1.1

(i) State the maximum speed of the train.

maximum speed = ... [1]

(ii) Describe the motion of the train between station A and station B.

.....

.....

..... [2]

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(iii) Calculate the distance between station A and station B.

distance = ... [3]

(iv) On a different day, the train takes 650 s to travel between station A and station B.

Suggest **one** change to the motion of the train that leads to this longer journey time.

.....

..... [1]

[Total: 8]

2 Table 2.1 contains information about the planet Mars.

Table 2.1

mass	6.4×10^{23} kg
gravitational field strength at surface	3.7 N / kg
average density	3900 kg / m^3

(a) Define gravitational field strength.

.....

..... [1]

(b) **(i)** An object has a weight of 42 N at the surface of the Earth.

Calculate the weight of the object at the surface of Mars.

weight = ... [2]

(ii) Calculate the volume of Mars.

volume = ... [2]

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(c) Fig. 2.1 shows a space buggy that is tested on Earth. The buggy is travelling at a constant speed in a straight line. The driving force on the buggy is 30 N.

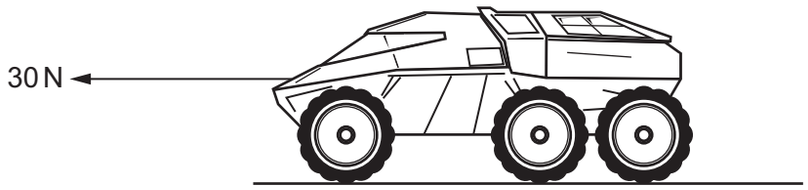


Fig. 2.1

(i) Draw and label **one** arrow on Fig. 2.1 to show the size and direction of the resistive forces on the buggy. [2]

(ii) Air resistance on Mars is less than air resistance on Earth.
The same driving force, 30 N, is exerted on the buggy on Mars.

1. State the effect this has on the resultant force on the buggy on Mars.

.....

2. State the relationship between resistive forces, driving force and resultant force.

.....
[1]

[Total: 8]

2 **(a)** A resultant force is applied to an object moving with a velocity v in a straight line.

(i) State **two** different changes to the motion that the resultant force may cause.

1 .....

2 .....
[2]

(ii) State **one** other way that forces may change a stationary object.

..... [1]

(b) Describe how a uniform metre ruler, a pivot and a selection of masses can be used to demonstrate that there is no resultant moment on an object in equilibrium.

You may include a labelled diagram in your answer.

.....

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.....

.....

..... [4]

[Total: 7]

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2 THERMAL PHYSICS

13 questions · Paper 2 & Paper 4

10 A hot gas and a cool liquid are at the same pressure.

Which row describes the separation and the motion of the particles of a hot gas compared to those of a cool liquid?

	separation of gas particles	motion of gas particles
A	greater than a cool liquid	faster than a cool liquid
B	greater than a cool liquid	slower than a cool liquid
C	less than a cool liquid	faster than a cool liquid
D	less than a cool liquid	slower than a cool liquid

12 The specific heat capacity of solid P is greater than that of solid Q.

What does this statement mean?

A Less energy is needed to raise the temperature by 1 °C of unit mass of solid P than unit mass of solid Q.

B Less energy is needed to melt unit mass of solid P than unit mass of solid Q.

C More energy is needed to raise the temperature by 1 °C of unit mass of solid P than unit mass of solid Q.

D More energy is needed to melt unit mass of solid P than unit mass of solid Q.

15 A student has a copper rod and a wooden rod. Both rods have the same dimensions.

The student puts one end of each rod into a beaker of crushed ice.

After a few minutes, the student picks up the rods by the ends **not** in the ice.

Which statement is correct?

A The copper rod feels colder than the wooden rod because copper is a better conductor than wood.

B The copper rod feels warmer than the wooden rod because copper is a better conductor than wood.

C The wooden rod feels colder than the copper rod because wood is a better conductor than copper.

D The wooden rod feels warmer than the copper rod because wood is a better conductor than copper.

11 A fixed mass of gas has a volume of 1.2 m^3 and is at a pressure of 200 kPa. The gas expands at constant temperature to a volume of 1.8 m^3.

What is the new pressure of the gas?

A 120 kPa **B** 130 kPa **C** 300 kPa **D** 600 kPa

13 When water evaporates, what escapes from the surface of the water?

A individual atoms

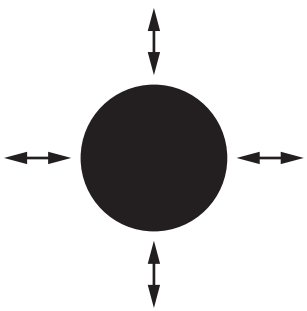
B individual molecules

C individual protons

D tiny drops of water

16 The diagram shows a black object X at constant temperature.

The arrows represent the absorption and emission of thermal radiation at the surface.



Object Y is at the same constant temperature as X and is identical except that it has a white surface.

Which row correctly compares object X with object Y?

	rate of absorption for X	rate of emission for X
A	higher	higher
B	lower	higher
C	higher	lower
D	lower	lower

14 A sealed container of gas is heated. The pressure of the gas increases.

Which statement explains this increase in pressure?

A The forces of the gas particles striking the walls of the container increase.

B The forces of attraction between the gas particles and the walls of the container increase.

C The gas particles collide with each other more frequently.

D The gas particles lose more energy when they strike the walls of the container.

12 The distance between two electricity pylons is 60 m. The cable that connects the two pylons is 62 m in length.

Why is the cable longer than the distance between the two pylons?

A to allow for contraction of the cable in cold weather

B to create a slope in the cable for electrons to flow down

C to keep the current low and the voltage high

D to reduce magnetic fields around the cable

14 Which statement about convection is **not** correct?

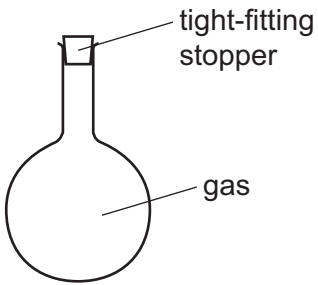
A It enables water in a pan on a cooker to get evenly heated.

B It happens in liquids and gases.

C It occurs because cooler liquids rise above warmer liquids.

D It occurs because the density of a liquid decreases when it is heated.

10 A flask with a tight-fitting stopper contains a gas. The gas and flask are initially at room temperature.



The gas and flask are cooled.

Which row describes the average speed of the gas particles and the pressure of the gas as the temperature decreases?

	average speed of the gas particles	pressure of the gas
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

4 Fig. 4.1 shows gas trapped in a cylinder by a piston.

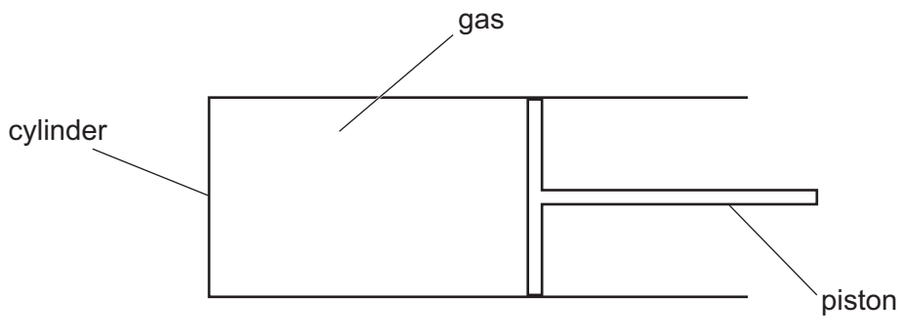


Fig. 4.1

(a) The volume of gas is 240 cm^3.
The piston is pushed to the left and is held in its new position.

(i) The pressure of the gas increases from 1.0×10^5 Pa to 1.4×10^5 Pa.

The temperature of the gas remains constant.

Calculate the volume of the gas when the piston is in its new position.

volume = ... cm^3 [3]

(ii) The area of the piston in contact with the gas is 1.9×10^{-3} m^2.

Calculate the force exerted on the piston by the gas when the piston is held in its new position.

force = ... [2]

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(iii) The distance moved by the piston is 0.036 m. The average force exerted by the piston as it moves is 220 N.

Calculate the mechanical work done by the piston. State the equation you use.

work done = ... [2]

(b) Explain, in terms of particles, why gases can be compressed but liquids cannot.

.....

.....

..... [1]

[Total: 8]

4 **(a)** A 12 V, 50 W immersion heater is used to heat 0.15 kg of water in a beaker. The water is initially at a room temperature of 20 °C. The specific heat capacity of water is 4200 J / (kg °C).

Calculate the energy supplied to raise the temperature of the water from 20 °C to 58 °C.

energy = ... [3]

(b) The immersion heater is removed from the beaker.

One metal rod and one plastic rod are placed in the beaker of hot water as shown in Fig. 4.1. The rods are at room temperature (20 °C) before they are placed into the beaker.

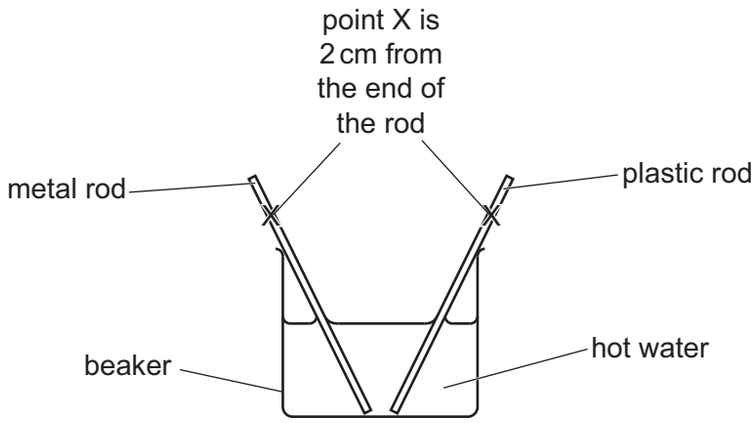


Fig. 4.1

Describe how the temperature of point X on each rod changes after the rods are placed in the beaker. Explain your answer.

.....

.....

.....

.....

..... [4]

[Total: 7]

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4 Fig. 4.1 shows a heater used to warm the air in a room.

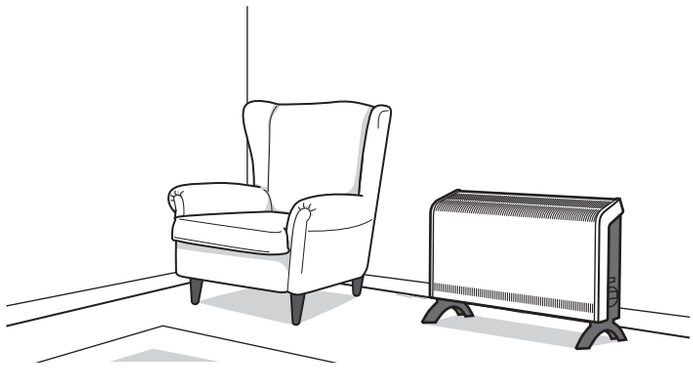


Fig. 4.1

(a) (i) State the main method of thermal energy transfer throughout the air in the room.

..... [1]

(ii) Explain how the heater warms **all** the air in the room.

.....

.....

.....

.....

..... [3]

(b) The power of the heater is 2.0 kW when it is connected to the mains supply with an e.m.f. of 230 V.

(i) Show that the current in the heater is approximately 8.7 A.

[2]

(ii) The plug connecting the heater to the mains supply is fitted with a fuse.

Fuse ratings of 3A, 5A, 10A and 13A are available.

State which fuse is used. Explain your answer.

fuse .....

explanation .....

.....

..... [2]

[Total: 8]

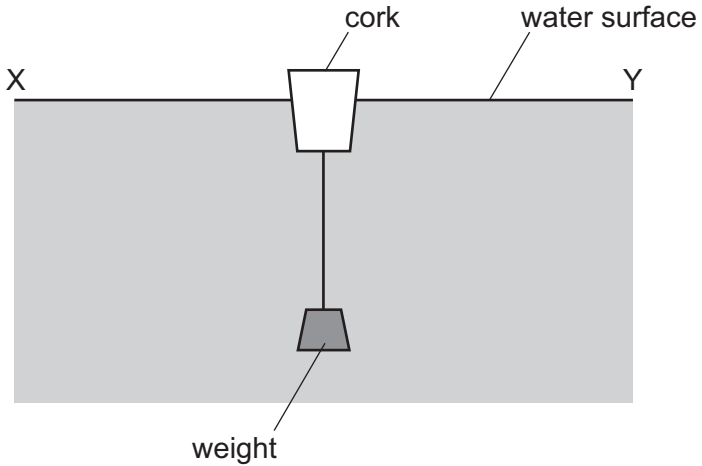
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3 WAVES

13 questions · Paper 2 & Paper 4

17 The diagram shows a cork with a weight attached so that the cork floats upright in water.



Transverse waves travel across the water from X to Y.

In which direction do the waves make the cork move?

A $\rightarrow \leftarrow$ right and left

B $\uparrow\downarrow$ up and down

C $\rightarrow$ only to the right

D $\leftarrow$ only to the left

18 A thin converging lens is used as a magnifying glass.

How does the image compare with the object?

A real and inverted

B real and upright

C virtual and inverted

D virtual and upright

22 Satellites are sometimes used to receive and re-transmit television signals.

Which row shows the type of satellite and the electromagnetic waves used?

	type of satellite	electromagnetic waves used
A	low orbit	infrared
B	low orbit	microwaves
C	geostationary	infrared
D	geostationary	microwaves

23 A sound wave travels through the air and strikes a solid block of steel.

Some of the sound is reflected from the steel, and some is transmitted through the steel.

How does the speed and the wavelength of the sound in steel compare to the speed and the wavelength of the sound in air?

	speed in steel	wavelength in steel
A	less than in air	less than in air
B	less than in air	greater than in air
C	greater than in air	less than in air
D	greater than in air	greater than in air

18 Parallel waves in a ripple tank are diffracted as they pass through a gap in a barrier.

Three changes that can be made to this arrangement are listed.

1 Decrease the wavelength of the waves.

2 Increase the amplitude of the waves.

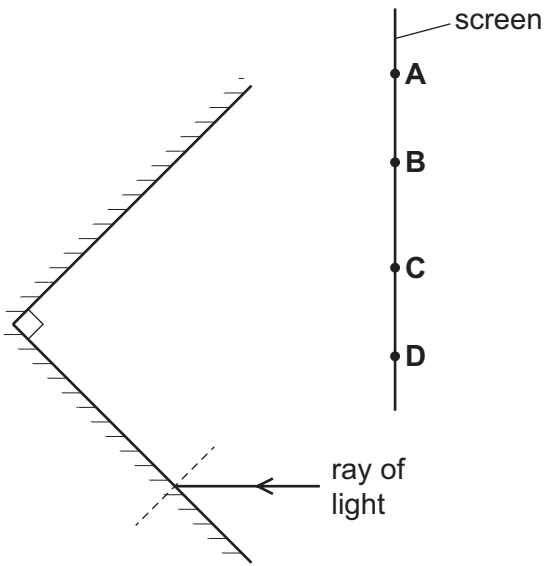
3 Decrease the size of the gap.

Which changes will cause the shape of the diffracted waves to be less curved?

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

19 The diagram shows two plane mirrors at 90° to each other. A ray of light is incident on one of the mirrors. The ray reflects off both mirrors before reaching a screen.

At which labelled point does the ray reach the screen?



22 Ultraviolet radiation and infrared radiation are both parts of the electromagnetic spectrum.

However, they differ in wavelength and frequency.

Which row describes how they differ?

	ultraviolet	infrared
A	longer wavelength than infrared	higher frequency than ultraviolet
B	longer wavelength than infrared	lower frequency than ultraviolet
C	shorter wavelength than infrared	higher frequency than ultraviolet
D	shorter wavelength than infrared	lower frequency than ultraviolet

23 A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does **not** change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

	amplitude	frequency
A	constant	constant
B	constant	varying
C	varying	constant
D	varying	varying

19 The frequency of a wave is doubled. The speed of the wave does **not** change.

What happens to the wavelength of the wave?

A It becomes four times as large.

B It does **not** change.

C It doubles.

D It halves.

20 The speed of light in a material is 50% of the speed of light in air.

What is the refractive index of the material?

A 0.5 **B** 2.0 **C** 6.0×10^8 **D** 1.5×10^8

7 **(a)** Fig. 7.1 is a scale drawing of light waves approaching a narrow slit.

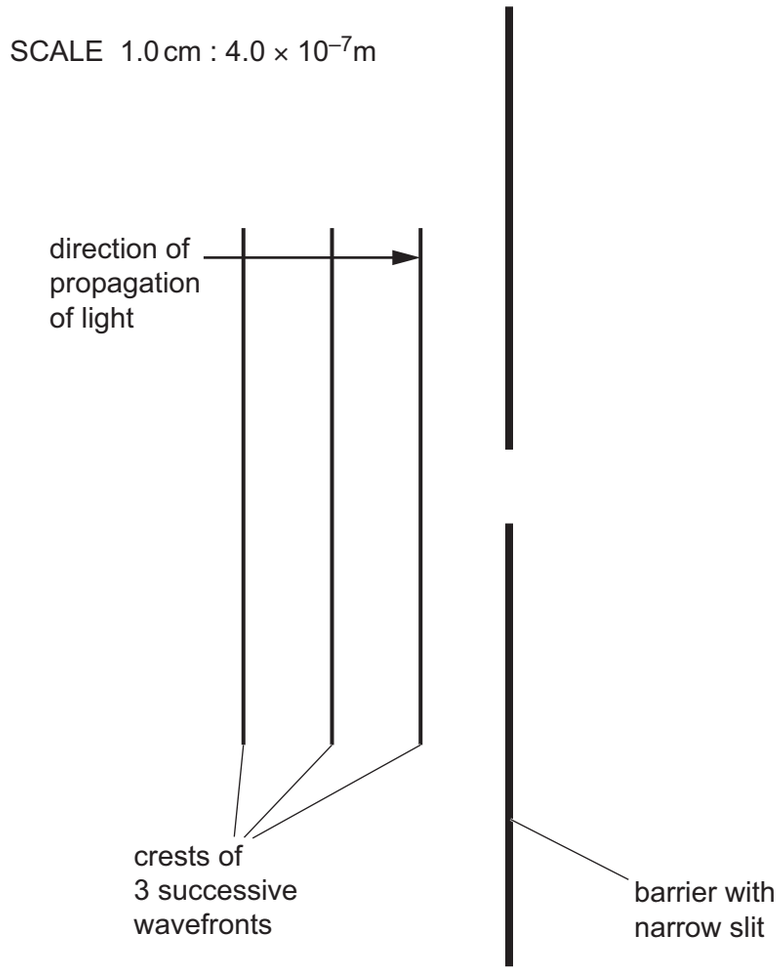


Fig. 7.1

(i) Name the wave effect produced by the narrow slit.

..... [1]

(ii) Using Fig. 7.1, determine the wavelength of the light. Give your answer to **two** significant figures.

wavelength = ... [2]

(iii) On Fig. 7.1, draw **three** wavefronts that have passed through the narrow slit. [3]

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(b) A foghorn emits a sound with frequency 380 Hz. The sound is heard by a ship 2.5 km away from the foghorn. The speed of sound in air is 330 m / s.

(i) Show that the wavelength of the sound is approximately 0.9 m.
State any equation you use in words or symbols.

[2]

(ii) Calculate the time it takes for sound to travel to the ship from the foghorn.

time = ... [2]

[Total: 10]

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5 A ray of light is incident on a soap film. Fig. 5.1 shows a magnified image of a small part of the soap film. The ray of light is refracted as it enters the soap film.

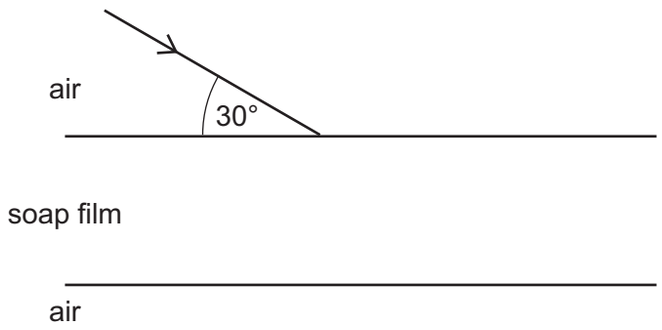


Fig. 5.1

The refractive index of the soap film is 1.28.

(a) Define refractive index in terms of the speed of light.

.....

..... [1]

(b) **(i)** Show that the angle of refraction as the light enters the soap film is approximately 43°.

[2]

(ii) On Fig. 5.1, carefully draw the refracted light ray in the soap film and label the angle of refraction. [2]

(c) The ray of light is monochromatic red light with a wavelength of 680 nm in air.

(i) Define monochromatic.

.....

..... [1]

(ii) Calculate the frequency of the light.

frequency = ... [3]

[Total: 9]

5 (a) (i) Electromagnetic waves have many uses.

On Fig. 5.1 draw **one** line from each use to the region of the electromagnetic (e-m) spectrum it uses.

Use	Region of e-m spectrum
security marking	gamma rays
Bluetooth	ultraviolet
optical fibres	infrared
detection of cancer	radio waves

Fig. 5.1

[2]

(ii) State **one** advantage of using microwaves compared with radio waves to transmit mobile (cell) phone signals.

.....

..... [1]

(b) Describe the difference between transverse waves and longitudinal waves.

.....

.....

.....

..... [2]

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(c) Fig. 5.2 shows a simplified diagram of seismic P-waves travelling through the Earth. The paths of the waves are curved.
Seismic waves are produced by rock movements in earthquakes.

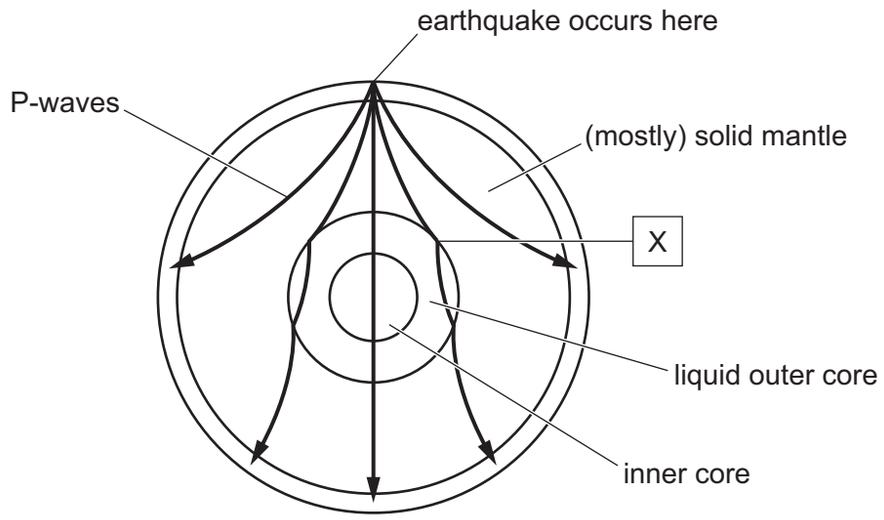


Fig. 5.2 (not to scale)

(i) State the type of wave that P-waves can be modelled as.

..... [1]

(ii) At point X, the P-wave travels from the solid mantle to the liquid core. There is a sudden change in direction of the path of the wave.

Explain the change in direction of the P-wave at point X.

.....

.....

..... [2]

(d) The Earth's surface absorbs incoming radiation from the Sun and also emits thermal radiation from its surface.

Over the past 50 years, scientists estimate that the average temperature of the Earth's surface has increased by approximately 0.75 °C.

Explain what is causing this average temperature rise.

.....

.....

.....

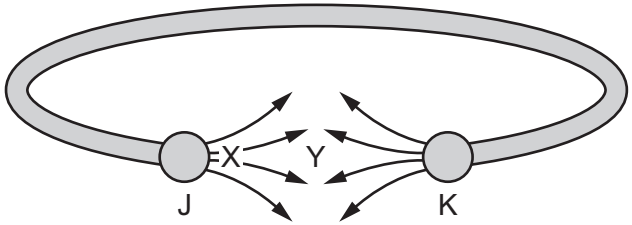
..... [2]

[Total: 10]

4 ELECTRICITY AND MAGNETISM

13 questions · Paper 2 & Paper 4

24 The diagram shows a bracelet containing two magnets. The arrows represent the pattern and the direction of the magnetic field due to the magnets.



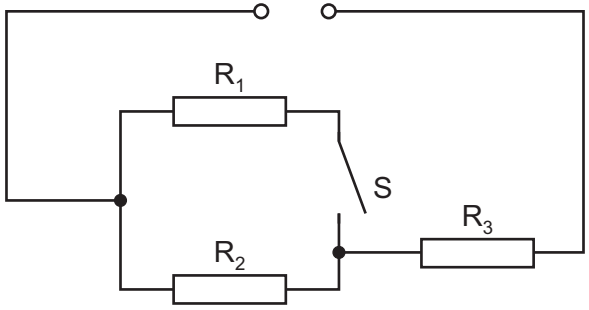
Which row identifies the magnetic pole at J, the magnetic pole at K and where the magnetic field is the strongest?

	magnetic pole at J	magnetic pole at K	where the magnetic field is the strongest
A	N	N	X
B	N	N	Y
C	S	S	X
D	S	S	Y

25 Which row describes an electric field and its direction?

	description of electric field	direction of electric field
A	region in which an electric charge experiences a force	direction of the force on a positive charge
B	region in which an electric charge experiences a force	direction of the force on a negative charge
C	region in which an electric charge produces a current	direction of the force on a positive charge
D	region in which an electric charge produces a current	direction of the force on a negative charge

27 The diagram shows three 20 Ω resistors, R_1, R_2 and R_3, and a switch S connected to a d.c. power supply.



Switch S is open.

The potential difference (p.d.) across R_3 is 9.0 V.

What is the p.d. across R_3 when switch S is closed?

A 3.0 V **B** 6.0 V **C** 12 V **D** 27 V

29 An electrical appliance has a symbol on it to indicate it is double-insulated.

Which statement is correct?

A A two-core, flexible cable with **no** earth wire is used to connect it to the mains supply.

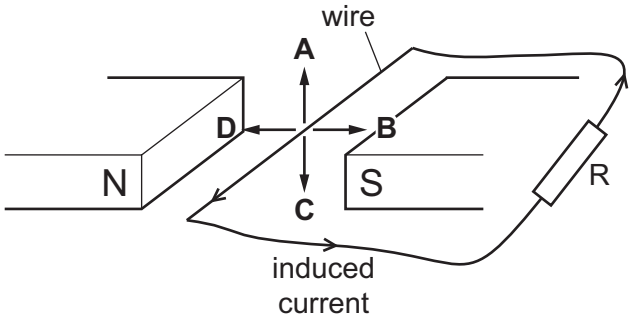
B It needs an earth wire that is connected to a fuse.

C **No** fuse or circuit breaker is needed.

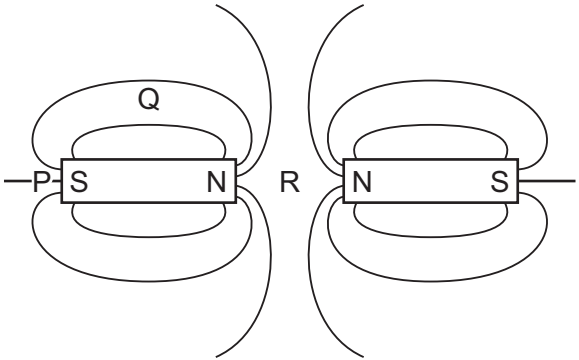
D The cable should be connected to the live and earth connectors of the plug but **not** the neutral.

29 A wire connected to a resistor R is moved in a magnetic field. A current is induced in the direction shown.

In which direction is the wire moved?



24 The diagram shows part of the magnetic field around two magnets.



Which list gives the order of magnetic field strengths at the positions P, Q and R, from weakest to strongest?

A $P \rightarrow Q \rightarrow R$

B $P \rightarrow R \rightarrow Q$

C $Q \rightarrow P \rightarrow R$

D $R \rightarrow Q \rightarrow P$

26 Copper wire is available in fixed lengths but in various diameters *d*. Each wire has a different resistance *R*.

Which relationship between *R* and *d* is correct?

A *R* is directly proportional to *d*.

B *R* is directly proportional to d^2.

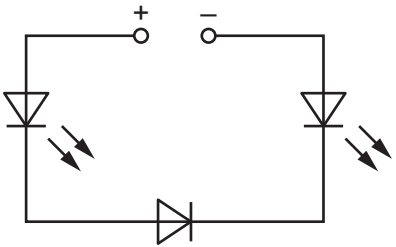
C *R* is inversely proportional to *d*.

D *R* is inversely proportional to d^2.

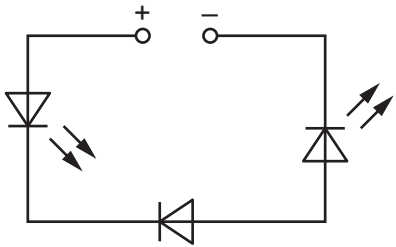
28 The circuits shown all include a d.c. power supply, one diode and two light-emitting diodes (LEDs).

In which circuit will both LEDs be turned on?

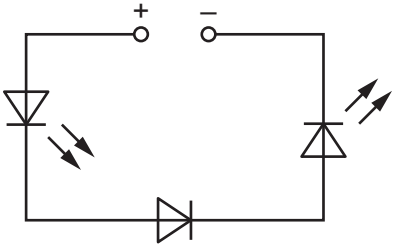
A



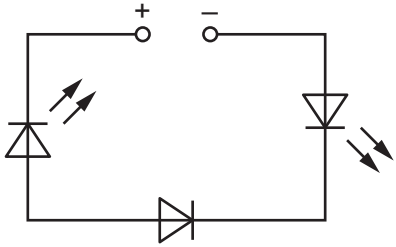
B



C



D



29 In which mains electric wires are the fuse and the switch placed?

	fuse	switch
A	earth	live
B	live	live
C	live	neutral
D	neutral	neutral

30 Which component is **not** found in an a.c. generator?

A coil of wire

B split-ring commutator

C magnetic poles

D slip rings

6 Fig. 6.1 shows a person using a magnetic window cleaner. The part on the outside of the window is attracted to the inside part through the glass window.

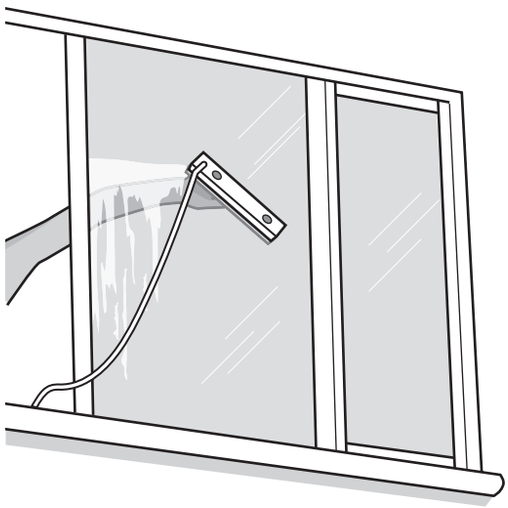


Fig. 6.1

Each part of the window cleaner contains two magnets. Fig. 6.2 shows the magnetic field between the parts of the window cleaner.

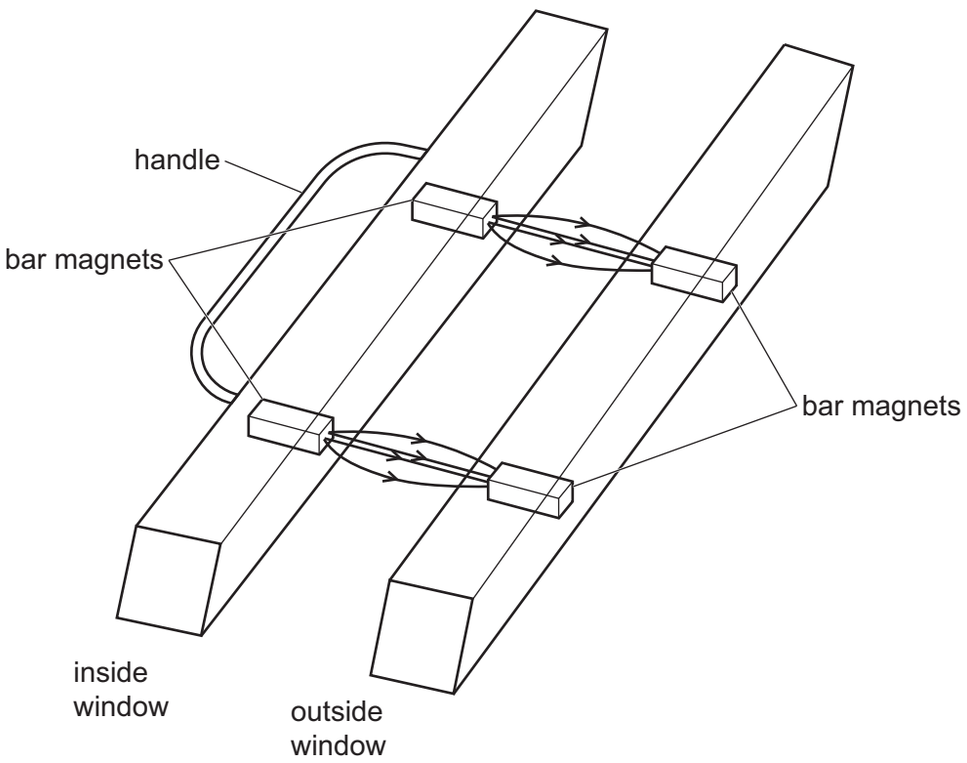


Fig. 6.2

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(a) Glass is **not** a magnetic material.

State the difference between magnetic and non-magnetic materials.

.....

..... [1]

(b) Suggest a suitable material for the magnets in the window cleaner. Explain your answer.

.....

..... [1]

(c) Label the poles of the magnets in Fig. 6.2. [1]

(d) State how the field lines in Fig. 6.2 show different strengths of the magnetic field between the magnets.

.....

..... [1]

[Total: 4]

* 0000800000006 *

3 Fig. 3.1 shows a mains electric heater used to heat a small room.

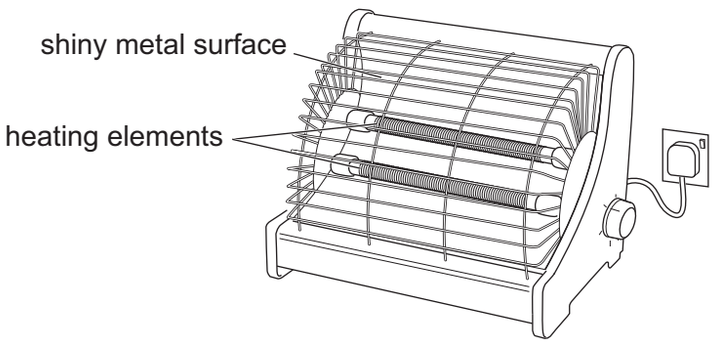


Fig. 3.1

(a) State the region of the electromagnetic spectrum which radiates thermal energy from the heater.

..... [1]

(b) Explain why the shiny metal surface behind the heating elements increases the thermal energy radiated into the room.

.....

.....

..... [2]

(c) The metal outer casing of the heater is earthed. State why this is an important safety feature.

.....

..... [1]

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(d) The mains voltage is 230 V. The two identical heating elements are connected in parallel. Each heating element has a resistance of 89 Ω.

(i) Calculate the current in **one** heating element.

current = ... [2]

(ii) Show that the electrical power of the heater is approximately 1200 W. State any equation you use in words or symbols.

[2]

(iii) The heater is 95% efficient at converting electrical work done to thermal energy.

Calculate the thermal energy emitted by the heater in **(d)(ii)** in 60 s. Give your answer to **two** significant figures.

thermal energy = ... [3]

[Total: 11]

7 **(a)** Fig. 7.1 shows a sketch of the current–voltage graph for an electrical component.

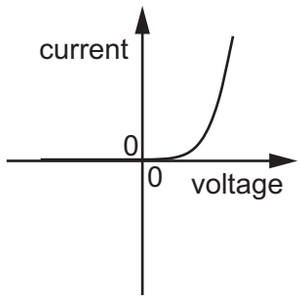


Fig. 7.1

(i) Name the electrical component.

Explain how you identified the component from the graph in Fig. 7.1.

name .....

explanation .....

.....

[2]

(ii) Draw the circuit symbol for this component.

[1]

(b) Fig. 7.2 shows an electric circuit for two identical electric heaters, A and B, connected to a mains supply of 230 V.

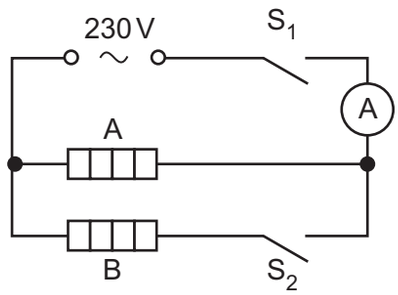


Fig. 7.2

S_1 is closed. S_2 is open. The reading on the ammeter is 3.9 A.

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(i) Calculate the resistance of heater A.

resistance = ... [2]

(ii) Calculate the energy transferred by heater A in 5.0 minutes.

energy = ... [3]

(iii) S_1 remains closed and S_2 is closed.

Determine the reading on the ammeter. Show your working.

ammeter reading = ... [2]

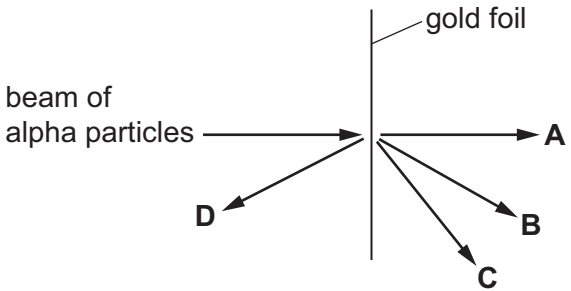
[Total: 10]

5 NUCLEAR PHYSICS

13 questions · Paper 2 & Paper 4

33 In an alpha particle scattering experiment, a beam of alpha particles is fired at a very thin piece of gold foil.

In which direction do most of the alpha particles travel after hitting the foil?



35 A detector placed near a radioactive isotope gives a measured count rate of 800 counts / minute. The half-life of the isotope is three hours.

The background count rate is 32 counts / minute.

What is the count rate on the detector after six hours?

A 192 counts / minute

B 200 counts / minute

C 224 counts / minute

D 232 counts / minute

34 An atom of boron contains 5 protons and 5 electrons. The nucleon number of the atom is 11.

How many neutrons are there in the atom?

A 5 **B** 6 **C** 11 **D** 16

36 Sodium-24 is a radioactive isotope that emits beta radiation.

Which nuclear equation shows how sodium-24 decays?

A $^{24}_{11}\text{Na} \rightarrow {}^{24}_{12}\text{Mg} + {}^{0}_{-1}\beta$

B $^{24}_{11}\text{Na} \rightarrow {}^{24}_{10}\text{Ne} + {}^{0}_{-1}\beta$

C $^{24}_{11}\text{Na} \rightarrow {}^{20}_{9}\text{F} + {}^{4}_{2}\beta$

D $^{24}_{11}\text{Na} \rightarrow {}^{20}_{13}\text{A}l + {}^{4}_{2}\beta$

33 $^{239}_{94}Pu$ is an isotope of plutonium.

The nucleus can absorb a neutron and then split to form two other nuclei.

Two nuclei that form are gold, $^{207}_{79}Au$, and phosphorus, $^{31}_{15}P$.

In this reaction, neutrons are also produced.

What is the number of neutrons produced when the nucleus splits?

A 1 **B** 2 **C** 4 **D** 5

37 Which safety precautions must be taken when using a source of gamma radiation?

1 Reduce the distance between the source and the person.

2 Reduce the time of exposure to the radiation.

3 Use a suitable shielding material between the source and the person.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

34 When a thin gold foil is bombarded with alpha particles, some alpha particles are deflected through large angles.

Which statement explains this deflection?

A Most of the atom consists of empty space.

B All of the positive charge and most of the mass of the gold atom are concentrated in a small volume.

C Positive charge in the gold atom is spread evenly throughout the atom.

D All of the negative charge is concentrated at its centre.

34 Three statements about alpha (α) particles and beta (β) particles are listed.

1 The magnitude of the charge on an α-particle is twice the charge on a β-particle.

2 The mass of an α-particle is large compared with the mass of a β-particle.

3 The kinetic energy of an α-particle is small compared to the kinetic energy of a β-particle.

Which statements help to explain the different ionising effects of α-particles and β-particles?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

33 The arrow shows the path of an α-particle travelling towards the centre of a gold nucleus.



Which description of the path of the α-particle after striking the gold nucleus is correct?

A The α-particle is deflected through a very small angle towards the top of the page.

B The α-particle is deflected back along its initial path.

C The α-particle carries on in a straight line with **no** deflection.

D The α-particle is captured by the nucleus and goes into orbit around it.

35 A beta-particle is a fast-moving electron.

Which statement explains how beta-particles are emitted from an atom?

A An electron is emitted as a beta-particle from an inner electron shell of the atom.

B An electron is emitted as a beta-particle from an outer electron shell of the atom.

C A neutron changes into a proton and a beta-particle is emitted from the nucleus.

D A proton changes into a neutron and a beta-particle is emitted from the nucleus.

9 **(a)** Describe the structure of an atom of helium-4, $^{4}_{2}He$.

.....

.....

.....

..... [3]

(b) The Sun is a medium-sized star powered by nuclear fusion reactions which release energy.

(i) State what happens during nuclear fusion reactions which form helium.

.....

.....

..... [2]

(ii) State **two** regions of the electromagnetic spectrum by which the Sun radiates most of its energy.

..... [2]

(c) Describe what happens to a star when most of the fuel in its centre has been converted to helium.

.....

.....

..... [2]

[Total: 9]

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9 Strontium-90 ($^{90}_{38}Sr$) is a radioactive isotope that contains 38 protons and 52 neutrons. Strontium-90 decays to form an isotope of yttrium (Y) by emitting beta (β) particles.

(a) (i) Suggest how the nucleus of a stable isotope of strontium differs from a nucleus of strontium-90. Explain your answer.

suggestion .....

explanation .....

.....

[2]

(ii) Complete the nuclide equation for the decay of strontium-90 to yttrium.

$$^{90}_{38}Sr \longrightarrow {}^{.....}_{.....}Y + {}^{.....}_{.....}\beta$$

[2]

(iii) Explain why scientists limit the amount of time they are exposed to radioactive strontium.

.....

..... [2]

(b) Yttrium is also unstable. A scientist places a sample of yttrium near a radiation detector. Table 9.1 shows the count rate recorded by the detector as the sample decays.

Table 9.1

time / h	recorded count rate / counts / min
0	68
50	49
100	38
150	32
200	26
250	24
300	20
350	21
400	20

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Fig. 9.1 shows a graph of the count rate **due to yttrium** against time.

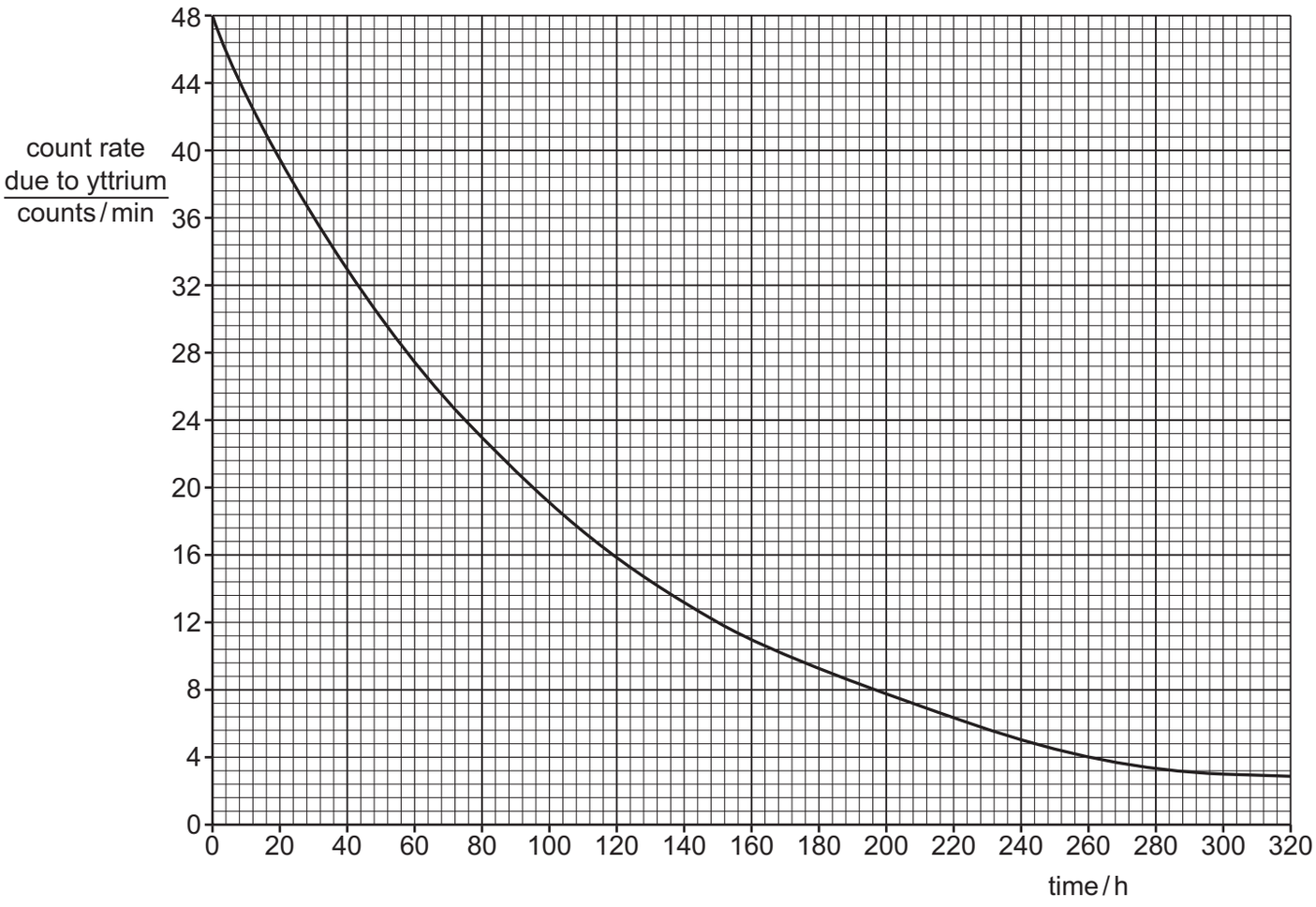


Fig. 9.1

(i) Use Fig. 9.1 to determine the half-life of yttrium. Show your working.

half-life = ... h [3]

(ii) Explain the difference between the count rate in Table 9.1 and the count rate due to yttrium plotted on the graph in Fig. 9.1.

.....

.....

..... [1]

[Total: 10]

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* 0000800000016 *

9 Strontium-90 is a radioactive isotope of strontium. The nuclide notation for strontium-90 is:

$^{90}_{38}Sr$

(a) (i) Explain what isotopes are.

.....

..... [1]

(ii) Complete Table 9.1 for strontium-90.

Table 9.1

particle	number in one atom	location
	38	outside nucleus
neutron		
	38	inside nucleus

[2]

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(b) Strontium-90 is used to measure the thickness of metal sheets in industry.
Strontium-90 decays by emitting beta (β) particles which pass through a metal sheet to a detector.

(i) One metal sheet is 0.75 mm thick. Suggest why strontium-90 is a suitable radioactive source to measure the thickness of the metal sheets.

.....

.....

..... [2]

(ii) The half-life of strontium-90 is approximately 27 years. Fig. 9.1 shows the shape of a decay curve.

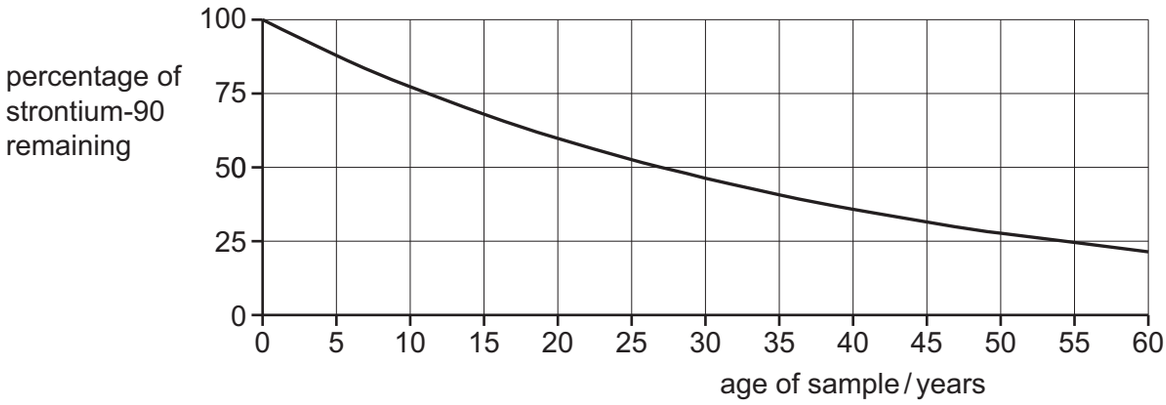


Fig. 9.1

The strontium-90 source is replaced with a new source after 15 years. Using Fig. 9.1, suggest why a strontium-90 source that is more than 15 years old needs to be replaced with a new source.

.....

..... [2]

[Total: 7]

6 SPACE PHYSICS

13 questions · Paper 2 & Paper 4

38 A student defines three periods of time:

- the time T_S taken for the Earth to orbit the Sun
- the time T_R taken for the Earth to rotate once on its axis
- the time T_M taken for the Moon to orbit the Earth.

Which row shows the shortest period of time and the longest period of time?

	shortest period of time	longest period of time
A	T_S	T_R
B	T_R	T_S
C	T_R	T_M
D	T_M	T_S

39 Which statement about the Universe is correct?

A The Universe consists of 100 000 galaxies.

B Redshift is a decrease in the observed wavelength of electromagnetic radiation emitted from receding galaxies.

C Light emitted from distant galaxies appears redshifted.

D Redshift of the light from distant galaxies gives evidence that the Universe is contracting.

38 A satellite is in orbit around the Earth. The orbit has radius r and period T.

Which expression gives the orbital speed of the satellite?

A $\frac{r}{2\pi T}$ **B** $\frac{T}{2\pi r}$ **C** $\frac{2\pi r}{T}$ **D** $\frac{2\pi T}{r}$

40 Which statement about the cosmic microwave background radiation (CMBR) is correct?

A CMBR now has a frequency higher than that of infrared radiation.

B CMBR has always had exactly the same frequency.

C CMBR had a shorter wavelength in the past.

D CMBR was produced a long time after the Universe was formed.

39 Comets move round the Sun.

Which statement describes the orbit of a comet?

A The orbit is circular with the Sun at the centre of the circle.

B The orbit is circular with the Sun **not** at the centre of the circle.

C The orbit is elliptical with the Sun at the centre of the ellipse.

D The orbit is elliptical with the Sun **not** at the centre of the ellipse.

40 A galaxy is at a distance of 60 million light-years from the Earth. The galaxy is moving away at a speed of 1300 km / s.

One light-year is about 9.5×10^{12} km.

There are about 3.2×10^{7} seconds in one year.

Which expression can be used to estimate the age of the Universe in years?

A $\dfrac{60 \times 10^{6} \times 3.2 \times 10^{7}}{1300 \times 9.5 \times 10^{12}}$

B $\dfrac{60 \times 10^{6} \times 9.5 \times 10^{12}}{1300 \times 3.2 \times 10^{7}}$

C $\dfrac{1300 \times 9.5 \times 10^{12}}{60 \times 10^{6} \times 3.2 \times 10^{7}}$

D $\dfrac{1300 \times 3.2 \times 10^{7}}{60 \times 10^{6} \times 9.5 \times 10^{12}}$

38 Which statement is correct?

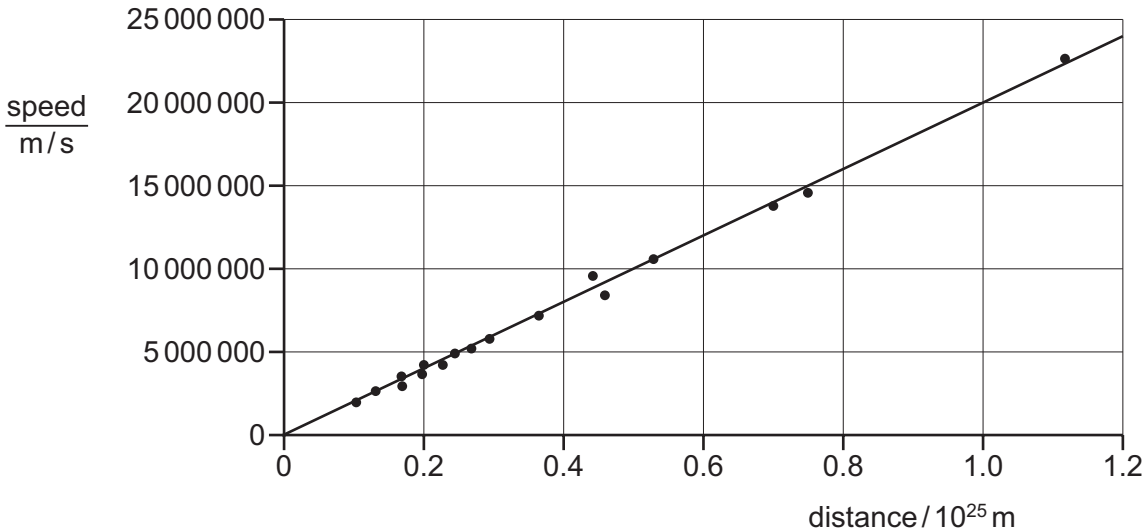
A The Earth orbits the Sun once in approximately 24 hours.

B The Earth orbits the Moon once in approximately 1 week.

C The Moon orbits the Earth once in approximately 1 month.

D The Sun orbits the Earth once in approximately 365 days.

40 A student plots a graph of the speed of different galaxies moving away from the Earth against their distance from the Earth.



Using the best straight line shown on the graph, what is the value of the Hubble constant H_0?

A $2.0 \times 10^{-18}\,s^{-1}$

B $2.2 \times 10^{-18}\,s^{-1}$

C $4.5 \times 10^{17}\,s$

D $5.0 \times 10^{17}\,s$

38 In which direction does the Sun appear to move every day, when viewed from the Earth?

A east to west

B north to south

C south to north

D west to east

39 Stars are powered by nuclear reactions that release energy.

Which statement describes the nuclear reaction in stable stars?

A Nuclear fission converts helium into hydrogen.

B Nuclear fission converts hydrogen into helium.

C Nuclear fusion converts helium into hydrogen.

D Nuclear fusion converts hydrogen into helium.

10 Jupiter and the Earth are planets in our Solar System.

(a) Describe the composition of Jupiter and the Earth.

Jupiter .....

the Earth .....
[2]

(b) The gravitational field strength at the surface of the Earth is approximately 9.8 N / kg. The gravitational field strength at the surface of Jupiter is approximately 23 N / kg.

(i) Define gravitational field strength.

.....

..... [2]

(ii) State **one** factor which causes the difference between the gravitational field strength at the surface of Jupiter and the gravitational field strength at the surface of the Earth.

.....

..... [1]

(c) State and explain the difference between the orbital speed of Jupiter and the orbital speed of the Earth.

statement .....

explanation .....

.....
[3]

[Total: 8]

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10 **(a)** Define a light-year in words.

..... [1]

(b) It takes light 490 s to travel from the Sun to the Earth.

Calculate the distance from the Sun to the Earth.

distance = ... [2]

(c) Fig. 10.1 is a scatter graph showing how the speed of galaxies moving away from the Earth varies with their distances from the Earth.

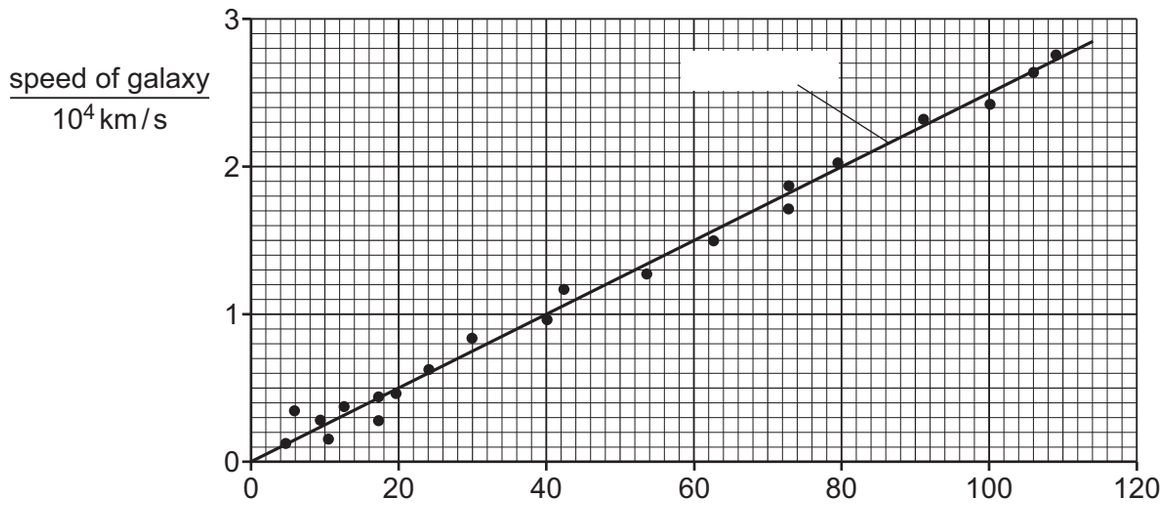


Fig. 10.1

A scientist draws a best-fit line on the scatter graph.

Use this best-fit line to determine a value for the Hubble constant.

Hubble constant = ... [3]

(d) The speed of a receding galaxy can be estimated using redshift.

Describe what is meant by redshift.

.....

.....

.....

..... [2]

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[Total: 8]

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10 Fig. 10.1 shows the orbit of the Earth and the orbit of a comet around the Sun.

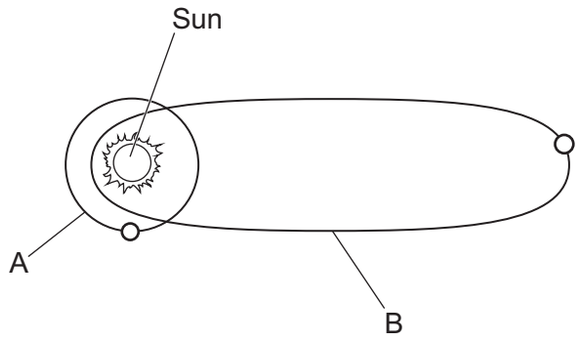


Fig. 10.1

(a) State which orbit, A or B, is the orbit of the comet.
Explain your answer.

orbit of comet is .....

explanation .....

.....
[1]

(b) Describe and explain how the motion of the comet changes as it orbits the Sun.

.....

.....

..... [2]

(c) At one position in its orbit, the comet is 6.6×10^{-6} light-years away from the Earth.

(i) State the meaning of light-year.

.....

..... [1]

(ii) Determine the distance in metres between the comet and the Earth.

distance = ... m [2]

[Total: 6]

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Answer Key

Answers — 1 MOTION, FORCES AND ENERGY

1.1 Physical quantities and measurement techniques

0625/21/O/N/25 (Q1): D
0625/22/M/J/25 (Q1): C
0625/22/O/N/25 (Q1): A
0625/23/M/J/25 (Q1): A
0625/23/O/N/25 (Q1): A
0625/23/O/N/25 (Q4): A

Answers — 1 MOTION, FORCES AND ENERGY

1.2 Motion

0625/21/M/J/25 (Q1): C
0625/21/O/N/25 (Q2): B
0625/22/F/M/25 (Q2): B
0625/22/F/M/25 (Q3): C
0625/22/M/J/25 (Q2): A
0625/22/M/J/25 (Q3): B
0625/22/O/N/25 (Q2): B
0625/23/M/J/25 (Q2): B
0625/23/M/J/25 (Q3): B
0625/23/M/J/25 (Q9): B
0625/23/O/N/25 (Q2): C
0625/23/O/N/25 (Q3): D

0625/41/M/J/25 Q1 — Train journey — vectors and speed–time graph [10 marks]

1(a) [1] (only) acceleration AND velocity circled
1(b)(i) [1] 56 m / s braille: 55 m / s
1(b)(ii) [2] accelerates OR speed increases AND (then) constant speed AND (then) decelerates OR speed decreases constant acceleration OR constant deceleration
1(b)(iii) [5] 26 000 m OR 26 km braille: 26 125 m (distance =) area under the speed–time graph OR (distance =) average speed ■ time taken ■ ■ 0.5 80 56 AND ■ 56 400 AND ■ ■ 0.5 50 56 OR ■ ■ ■ + ■ 1 400 530 56 2 braille: 0.5 ■ 100 ■ 55 AND 55 ■ 400 AND 0.5 ■ 50 ■ 55 OR 1 {400 + 550} 55 2 ■ ■
1(b)(iv) [1] any one from: • lower acceleration OR less acceleration • lower deceleration OR less deceleration • lower maximum speed OR lower average speed OR lower constant speed 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 9 of 16

0625/41/O/N/25 Q1 — Train — velocity, kinetic energy and braking [12 marks]

1(a) [1] the velocity is the speed in a particular direction OR velocity has a direction OR velocity is 56 m / s in a certain direction velocity is a vector OR speed is a scalar
1(b) [3] 6.9 ■ 108 J OR 690 000 000 J OR 690 MJ (KE =) ½ mv2 OR 0.5 ■ 440 000 ■ (56)2
1(c)(i) [3] 5.3 ■ 105 N OR 530 000 N (F =) ma OR 440 000 ■ 1.2
1(c)(ii) [3] 1300 m OR 1.3 km
1(c)(ii) [2] (d =) W / F OR (Δ)t = Δv / a OR 56 / 1.2 OR (Δ) t = change of momentum / force (distance =) 6.9 ■ 108 / 530 000 OR (distance =) average velocity ■ time taken OR 1.3 ■ 10N

0625/42/F/M/25 Q4 — Train — acceleration, time and force [10 marks]

4(a)(i) [1] (acceleration is) rate of change in velocity OR (acceleration is) change in velocity per unit time OR (acceleration is) change in velocity per second
4(a)(ii) [2] 200 km = 200 000 m OR ■ 1000 seen division by {60 ■ 60} seen OR division by 3600 seen
4(a)(iii) [3] 80 s OR 79 s (t =) Δv / a OR 56 / 0.7(0)
4(b)(i) [3] 310 000 N OR 5 3.1 1 0 N ■ F = ma OR ( ) 440 000 0.7 0 ■
4(b)(ii) [1] (statement:) reduces acceleration OR lower (maximum) velocity AND (explanation:) resultant/net force decreases

0625/42/M/J/25 Q1 — Ice skater — speed–time graph and deceleration [8 marks]

1(a) [1] constant speed OR uniform speed
1(b) [3] 33 m (distance =) area under graph OR bh OR speed ■ time OR 11 ■ 3
1(c)(i) [1] (rate of) decrease of velocity OR decrease in velocity (per unit time) OR negative acceleration
1(c)(ii) [3] tangent drawn at 9.0 s on graph 0.50–0.70 m / s2 correct method of gradient calculation using any two correct pairs of co-ordinates (± ½ a small square) from their tangent 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 10 of 19

0625/43/M/J/25 Q1 — Car braking — speed–time graph and deceleration [11 marks]

1(a) [1] 11.5 m / s
1(b) [5] 45 m (distance =) area under graph OR ½ bh (+ bh) {0.5 ■ 20} + ½ {20 ■ 3.5} OR {10 + 35} OR

[{0.5 + 4} ÷ 2] ■ 20
1(c) [5] 5.7 m / s2 (deceleration =) (■)v ÷ ■t OR deceleration = gradient (0 –) 20 ÷ {4(.0) - 0.5} OR 2 ÷ 3.5

0625/43/O/N/25 Q1 — Dropped ball — speed–time graph, energy and air resistance [12 marks]

1(a)(i) [1] constant acceleration
1(a)(ii) [2] area under the graph OR ½ bh OR average speed ■ time 0.5 ■ 5.9 ■ 0.6 OR 1.77 (to ■ 3 sf)
1(a)(iii) [3] 9.5 J (■Ep =) mg(■)h OR 0.54 ■ 9.8 ■ 1.8
1(a)(iv) [1] candidate's 1(a)(iii) value ■ 0.5 (2sf)
1(b) [3] any three from: • (initially there is acceleration due to) an unbalanced force OR resultant force OR downward force OR weight OR gravitational force acting on the ball • as speed increases the air resistance increases • (as air resistance increases) resultant force downwards decreases OR acceleration decreases • (eventually) ball falls with constant speed when air resistance = weight OR ball falls with terminal velocity when air resistance = weight
1(c) [2] (some of the) energy is: transferred to the thermal energy (store) of the ground (when ball hits ground) OR converted / transferred to thermal energy (store) of the air OR used to deform ball / transferred to the elastic energy (store) of the ball (as it contacts the ground) gravitational potential energy (in the store at C) will be less (than at A) OR (ball has) less kinetic energy (in its store just after B) 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 12 of 20

Answers — 1 MOTION, FORCES AND ENERGY

1.3 Mass and weight

0625/21/M/J/25 (Q3): D
0625/21/O/N/25 (Q3): B
0625/22/F/M/25 (Q4): D
0625/22/F/M/25 (Q39): C
0625/22/M/J/25 (Q4): D
0625/22/O/N/25 (Q3): C
0625/22/O/N/25 (Q38): A
0625/23/M/J/25 (Q4): D

0625/41/O/N/25 Q2 — Mars — gravitational field strength, weight and density [10 marks]

2(a) [1] (gravitational field strength is) force per unit mass (on an object in a gravitational field)
2(b)(i) [3] 16 N W = mg OR (m =) W÷g OR 42 / 9.8 OR (mass of object =) 4.3 (kg)
2(b)(ii) [3] 1.6 ■ 1020 m3 (V =) m / ρ OR (V =) 6.4 ■ 1023 / 3900
2(c)(i) [1] arrow parallel to driving force AND pointing to the right 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 12 of 18
2(c)(i) [1] (arrow pointing to the right) labelled 30 N
2(c)(ii) [1] 1 (resultant force) increases OR there is a resultant force (in the direction of the driving force) OR 2 resultant force = driving force – resistive force(s)

Answers — 1 MOTION, FORCES AND ENERGY

1.4 Density

0625/21/M/J/25 (Q4): C
0625/21/O/N/25 (Q4): B
0625/22/M/J/25 (Q5): C
0625/22/O/N/25 (Q4): C
0625/23/M/J/25 (Q5): A
0625/23/O/N/25 (Q5): D

Answers — 1 MOTION, FORCES AND ENERGY

1.5 Forces

0625/21/M/J/25 (Q2): B
0625/21/M/J/25 (Q5): B
0625/21/M/J/25 (Q6): D
0625/21/M/J/25 (Q8): A
0625/21/O/N/25 (Q5): C
0625/21/O/N/25 (Q6): D
0625/22/F/M/25 (Q1): B
0625/22/F/M/25 (Q5): A
0625/22/M/J/25 (Q6): A
0625/22/O/N/25 (Q5): A
0625/23/M/J/25 (Q6): B
0625/23/M/J/25 (Q7): B
0625/23/O/N/25 (Q6): B
0625/23/O/N/25 (Q7): B

0625/41/M/J/25 Q2 — Resultant force and moments in equilibrium [7 marks]

2(a)(i) [2] direction (changes) any one from: • magnitude of the velocity (changes) • speed (changes) • (there is) acceleration
2(a)(ii) [1] any one from: • (change) size • (change) shape
2(b) [4] 1 place (the centre of) the metre ruler on the pivot owtte OR (labelled) diagram showing metre ruler on a pivot 2 add mass on one side (of pivot) and then add mass on other side to balance the ruler owtte 3 (moment =) force ■ perpendicular distance (from pivot) 4 sum of clockwise moments = sum of anticlockwise moments (when in equilibrium)

0625/42/F/M/25 Q1 — Spring — spring constant and the limit of proportionality [7 marks]

1(a) [3] 50 000 N / m OR 5.0 ■ 104 N / m (k =) F ■ x OR k is (determined from) the gradient (of the line in Fig. 1.1)
1(b) [1] Any one from: • The graph is a straight line (through the origin) • The slope / gradient is constant • The graph / line does not curve
1(c) [2] 1300 N any negative number OR indication that force is in opposite direction
1(d) [1] (Force is a) vector quantity. Forces have (both magnitude / size and) direction

0625/42/M/J/25 Q2 — Balanced ruler — moment, centre of gravity and density [8 marks]

2(a) [1] (moment =) force ■ perpendicular distance (from the pivot)
2(b) [1] X AND 50 cm labelled to the right of the pivot AND left of 80 cm
2(c)(i) [3] sum of clockwise moments = sum of anticlockwise moments {mg ■ 8} + {0.12 ■ 38} = {0.34 ■ 32} any one from: • mg = [{0.34 ■ 32} – {0.12 ■ 38}] ÷ 8 • mg = {10.88 – 4.56} ÷ 8 • mg = 6.32 ÷ 8 • mg = 0.79 • (mass =) 0.79 ÷ 9.8
2(c)(ii) [3] 520 kg / m3 (density =) mass ÷ volume OR 0.081 ÷ {1(.0) ■ 2.6 ■ 10–2 ■ 6(.0) ■ 10–3}
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0625/42/O/N/25 Q2 — Truck — resultant force and circular motion [10 marks]

2(a)(i) [5] 7400 N a = Δv / Δt OR a = {22 – 13} / 4.5 OR 2.0 (m / s2) SEEN (F =) ma OR 3700 ■ 2.0
2(a)(ii) [2] air resistance OR drag friction (between tyres and road)
2(b)(i) [1] arrow from car pointing towards the centre of the circle
2(b)(ii) [2] any two from: • car has higher speed • less friction (between car and track) OR not enough friction • larger mass 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 13 of 20

Answers — 1 MOTION, FORCES AND ENERGY

1.6 Momentum

0625/21/M/J/25 (Q7): C
0625/21/O/N/25 (Q7): A
0625/22/F/M/25 (Q6): C
0625/22/F/M/25 (Q8): A
0625/22/M/J/25 (Q7): D
0625/22/O/N/25 (Q6): C
0625/23/M/J/25 (Q8): B
0625/23/O/N/25 (Q8): B

0625/42/F/M/25 Q2 — Trolley collision — conservation of momentum [10 marks]

2(a) [1] (statement momentum is) zero / 0 AND (explanation) momentum is mass ■ velocity OR the velocity (of B) is zero
2(b) [5] 1.1 kg Conservation of momentum OR momentum after (collision) = momentum before (collision) OR {0.75 ■ 0.44} = {0.75 + m} ■ 0.18 momentum before collision = 0.75 × 0.44 (+ m 0 ■ = 0.33) OR momentum after collision = (0.75 + m) ■ 0.18
2(c)(i) [2] 0.13 N (F =) Δp ■ (Δ)t OR (0.75 + 1.1) ■ 0.18 ■ 2.6 OR (F =) 0.33 ■ 2.6 ( )■ ■ 0.75 0.18 2.6 F = + ■ ■
2(b) [1] 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED February/March 2025 © Cambridge University Press & Assessment 2025 Page 7 of 14
2(c)(ii) [1] (time taken) Increases AND use of F(Δ)t = Δ{mv} OR F ■1 / (Δ)t OR same momentum change required

0625/42/M/J/25 Q3 — Toy train collision — momentum and force [11 marks]

3(a) [1] (momentum =) mass ■ velocity OR (p =) mv
3(b) [7] (velocity =) 0.19 m / s AND (direction) to the right (momentum before collision) = {0.45 ■ 0.34} – {0.21 ■ 0.12} OR 0.1278 (momentum after collision) = 0.66 ■ v momentum before collision = momentum after collision OR {0.45 ■ 0.34} – {0.21 ■ 0.12} = 0.66 ■ v OR 0.153 – 0.0252 = 0.66 ■ v
3(c) [3] (–) 0.12 N FΔt = Δ{mv} OR (F =) Δp ÷ (Δ)t OR (F =) Δ{mv} ÷ (Δ)t OR (F =) 0.26 ÷ 2.1 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 12 of 19

0625/42/O/N/25 Q1 — Tennis ball — scalars, vectors and momentum [9 marks]

1(a)(i) [1] scalar has magnitude (size) only OR vector has (magnitude and) direction
1(a)(ii) [2] mass, energy and temperature all correct and no additional quantities 2 marks 2 correct and no additional quantities 1 mark
1(b)(i) [3] decrease in kinetic energy = increase in gravitational (potential) energy OR ½ mv2 = mg(■)h v2 = 2gh OR v2 = 2 ■ 9.8 ■ 5.0 OR v2 =98 v = √98 OR 9.899
1(b)(ii) [3] 0.57 kg m / s (upwards) OR 0.57 Ns (upwards) (momentum =) mv OR 5.8 ■ 10-2 ■ 9.9 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 12 of 20

0625/43/M/J/25 Q2 — Space vehicle — momentum and terminal velocity [9 marks]

2(a) [1] (p =) mv OR mass ■ velocity
2(b) [5] 7790 m / s momentum before (collision) = momentum after (collision) OR mcvc + mnvn = 6.55 ■ 106 OR (momentum of cone =) 120 ■ 7850 OR 9.42 ■ 105 720v + {120 ■ 7850} = 6.55 ■ 106 OR (momentum after collision =) 6.55 ■ 106 – 9.42 ■ 105 OR 5.6 ■ 106
2(c) [3] 12 000 kg 120 000 (N) OR (m =) W ÷ g OR (m =) 1.2 ■ 10N ÷ 9.8 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 9 of 13

0625/43/O/N/25 Q2 — Cricket catch — change in momentum and force [8 marks]

2(a) [3] (–) 3.5 kgm / s (■p =) m(■)u OR m{v – u} OR 0.14 ■ {25 (– 0)}
2(b) [5] 360 N 0.0097 (s) OR 9.7 ■ 10–3 (s) OR 3.6 ■ 10n (F =) (■)p / t OR 3.5 / 0.0097

Answers — 1 MOTION, FORCES AND ENERGY

1.7 Energy, work and power

0625/21/O/N/25 (Q8): C
0625/22/F/M/25 (Q9): D
0625/22/F/M/25 (Q10): B
0625/22/F/M/25 (Q12): D
0625/22/M/J/25 (Q8): C
0625/22/M/J/25 (Q9): D
0625/22/O/N/25 (Q7): D
0625/22/O/N/25 (Q8): D
0625/23/O/N/25 (Q9): C
0625/23/O/N/25 (Q10): D

0625/41/M/J/25 Q3 — Skateboarder — energy transfer and work done [8 marks]

3(a) [2] gravitational potential to kinetic to gravitational potential to thermal (store) OR to internal (store)
3(b)(i) [3] 1800 J (W =) Fd OR (W =) ■ ■ 65 9.8 2.8 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 10 of 16
3(b)(ii) [3] F = Δp ■(Δ)t AND Δp is change in momentum, (Δ)t is time (taken) OR F = Δ{mv} ■(Δ)t AND Δ{mv} is change in momentum, (Δ)t is time (taken) OR force = rate of change in momentum OR force = change in momentum divided by time (taken) F = Δp ■(Δ)t OR F = Δ{mv} ■(Δ)t OR F = I ■(Δ)t

0625/41/O/N/25 Q3 — Solar cell — energy transfer and efficiency [12 marks]

3(a) [2] (electromagnetic) radiation / light (from the Sun) (produces) electrical (work done)
3(b) [1] (black) is a good absorber / poor reflector (of radiation) owtte
3(c)(i) [3] 0.54 kW OR 540 W (output power =) total power input ■ efficiency (÷100) OR (output power =) 0.72 ■ 75 ÷ 100 OR 5.4 ■ 10N
3(c)(ii) [1] the amount of (electrical) energy transferred by a 1 kW appliance in 1 hour owtte OR energy transferred in one hour at a rate of transfer of 1 kW
3(c)(iii) [4] 7 Any one from: • E = Pt • energy produced by one cell per year OR 3(c)(i) ■ 6 ■ 365 • total power output required OR 7400 ÷ {365 ■ 6} • household energy used per day OR 7400 ÷ 365 • energy produced by one cell per day OR 3(c)(i) ■ 6 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 13 of 18
3(c)(iii) [1] Any one from: • household energy used per year ÷ energy produced by one cell per year • total power output required ÷ power output of one cell • household energy used per day ÷ energy produced by one cell per day

0625/42/O/N/25 Q3 — Solar panels — efficiency, a.c. and d.c. [14 marks]

3(a) [2] more electricity generated OR makes solar panels more efficient black is a good absorber OR black is a poor reflector owtte
3(b) [5] 22 kW OR 22 000 W (total power input =) {useful power output (■ 100%)} / (%) efficiency OR (total power input =) {3.5 ■ 100} / 16 OR (total power input =) 3.5 / 0.16 (total power input =) 3.5 / 0.16 OR (total power input =) {3.5 ■ 100} / 16
3(c) [1] alternating current reverses direction OR direct current is only in one direction
3(d) [1] electricity can be used when there is no Sun OR when it is dark or cloudy OR at night
3(e) [5] 29 000 C (Q =) I t OR (Q =) 4.0 ■ 2.0 ■ 60 ■ 60 correct conversion from h to s SEEN 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 14 of 20

0625/43/M/J/25 Q3 — Archer's bow — work, elastic energy and arrow speed [9 marks]

3(a) [2] (F =) W ÷ d OR W = Fd 110 ÷ 0.45 OR 244 (N)
3(b)(i) [5] 86 m / s Ek = ½ mv2 OR 110 = ½ mv2 OR 110 = ½ ■ 0.03 ■ v2 (v =) √ [{2 ■ 110} ÷ 0.03] OR v2 = {2 ■ 110} ÷ 0.03 OR v2 = 7333
3(b)(ii) [2] air resistance / drag / friction energy lost to surroundings / energy transferred to thermal / internal energy

0625/43/O/N/25 Q3 — Coal-fired power station — Sankey diagram and efficiency [7 marks]

3(a)(i) [2] (wasted power =) 2200 + 40 + 25 OR 2265 OR (useful power output =) power input – was
power 1140 (MW) (■ 3 sf)
3(a)(ii) [3] 0.32 OR 32% (efficiency =) {useful power output / total power input} (■ 100%) OR 1100 /
3400
3(b) [2] water is pumped into the ground and heated by hot rocks OR heat from hot rocks undergrou
used to turn water into steam steam used to heat buildings OR steam used to turn generator / turbi
produce electricity 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 20
University Press & Assessment 2025 Page 13 of 20

Answers — 1 MOTION, FORCES AND ENERGY

1.8 Pressure

0625/21/M/J/25 (Q9): D

0625/21/O/N/25 (Q9): A

0625/22/F/M/25 (Q7): B

0625/22/O/N/25 (Q9): B

0625/43/O/N/25 Q4 — Liquid pressure, density and thermal expansion [19 marks]

4(a)(i) [1] 930 kg / m3

4(a)(ii) [14] 1400 N (Δ)p = ρg(Δ)h OR 930 ■ 9.8 ■ 1.1 OR 10 000 (N / m2) area = length ■ width OR 0.45 ■ 0.30 OR 0.14 (m2) (F=) p ■ A OR 10 000 ■ 0.14 Alternative method: 1400 N volume (of liquid above block) = length ■ width ■ height OR 0.45 ■ 0.30 ■ 1.1 OR 0.15 (m3) mass (of liquid above block) = ρ ■ v OR 930 ■ 0.15 OR 140 (kg) (F =) mg OR 140 ■ 9.8

4(b) [2] thermal expansion OR to prevent buckling / bending when heated any one from: • particles vibrate faster • particles have greater speed / kinetic energy / momentum • vibrations (of particles) take up more space • (average) space / separation between particles increases

4(c) [2] (plastic is a good) insulator / poor conductor (handle) will not get too hot to touch OR reduces heat flow to hand 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 14 of 20

Answers — 2 THERMAL PHYSICS

2.1 Kinetic particle model of matter

0625/21/M/J/25 (Q10): A
0625/21/M/J/25 (Q11): B
0625/21/M/J/25 (Q14): A
0625/21/O/N/25 (Q10): A
0625/21/O/N/25 (Q11): C
0625/22/F/M/25 (Q13): A
0625/22/F/M/25 (Q16): B
0625/22/M/J/25 (Q10): B
0625/22/M/J/25 (Q11): A
0625/22/M/J/25 (Q12): C
0625/22/O/N/25 (Q10): D
0625/23/M/J/25 (Q10): D
0625/23/M/J/25 (Q11): A
0625/23/M/J/25 (Q12): D
0625/23/O/N/25 (Q11): A
0625/23/O/N/25 (Q12): B

0625/42/O/N/25 Q4 — Gas in a cylinder — pressure, volume and work [11 marks]

4(a)(i) [5] 170 (cm3) pV = constant OR 1.0 ■ 105 ■ 240 = 1.4 ■ 105 ■ V2 (V2 =) {1.0 ■ 105 ■ 240} / 1.4 ■ 105 OR 1.7 ■ 10N

4(a)(ii) [3] 270 N (F =) pA OR (F =) 1.4 ■ 105 ■ 1.9 ■ 10-3

4(a)(iii) [2] (W =) Fd in words or symbols 7.9 J

4(b) [1] distance between gas particles is much larger (than distance between liquid particles) OR particles in liquid are touching AND particles in gas are far apart owtte 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 15 of 20

Answers — 2 THERMAL PHYSICS

2.2 Thermal properties and temperature

0625/21/M/J/25 (Q12): C
0625/21/M/J/25 (Q13): B
0625/21/O/N/25 (Q12): A
0625/21/O/N/25 (Q13): A
0625/22/F/M/25 (Q14): D
0625/22/F/M/25 (Q15): A
0625/22/F/M/25 (Q17): C
0625/22/M/J/25 (Q13): A
0625/22/M/J/25 (Q14): B
0625/22/O/N/25 (Q11): D
0625/22/O/N/25 (Q12): C
0625/22/O/N/25 (Q13): B
0625/22/O/N/25 (Q14): D
0625/23/M/J/25 (Q13): D
0625/23/M/J/25 (Q14): A
0625/23/O/N/25 (Q13): C
0625/23/O/N/25 (Q14): A

0625/41/O/N/25 Q4 — Immersion heater and conduction — specific heat capacity [10 marks]

4(a) [5] 24 000 J OR 2.4 ■ 104 J (ΔE =) mcΔ■ OR (ΔE =) 0.15 ■ 4200 ■ {58 – 20} (Δ■ =) 58 – 20 OR 38 (°C) seen

4(b) [5] (temperature of point X) on metal rod increases faster ORA (temperature of point X) increases (with time) thermal energy is transferred) by conduction metal rod (transfers thermal energy through movement of) delocalised / free electrons. ORA

0625/42/O/N/25 Q5 — Evaporation and braking distance [5 marks]

5(a)(i) [2] particles with more (kinetic) energy escape OR fast(er) particles escape (particles) escape from the surface

5(a)(ii) [2] windy OR (air) temperature higher / Sun shining particles that escape from surface blown away (and unable to return) OR more energy given to the particles (to escape)

5(b) [1] braking distance increases AND friction (between tyres and road) decreases 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 16 of 20

0625/43/M/J/25 Q4 — Copper pan — conduction and specific heat capacity [11 marks]

4(a) [3] any three from: • (copper / metal contains) free / delocalised electrons • electrons carry (thermal) energy through metal • electrons collide with (distant) ions • lattice vibrations transfer energy (to neighbouring ions) OR ions vibrate and cause (nearby / adjacent) ions to vibrate

4(b)(i) [3] energy transferred per unit mass per unit temperature change (thermal) energy (transferred) per unit temperature change

4(b)(ii) [5] 0.51 J / (g ■C) (energy lost by metal =) 54 ■ c ■ 69 OR (energy gained by water =) 50 ■ 4.2 ■ 9 OR 1890 energy lost by metal = energy gained by water OR 54 ■ c ■ 69 = 50 ■ 4.2 ■ 9 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 10 of 13

Answers — 2 THERMAL PHYSICS

2.3 Transfer of thermal energy

0625/21/M/J/25 (Q15): A
0625/21/M/J/25 (Q16): A
0625/21/O/N/25 (Q14): C
0625/21/O/N/25 (Q15): A
0625/21/O/N/25 (Q16): C
0625/22/F/M/25 (Q11): A
0625/22/F/M/25 (Q18): D
0625/22/M/J/25 (Q15): A
0625/22/M/J/25 (Q16): B
0625/22/O/N/25 (Q15): A
0625/22/O/N/25 (Q16): A
0625/23/M/J/25 (Q15): C
0625/23/M/J/25 (Q16): B
0625/23/O/N/25 (Q15): A
0625/23/O/N/25 (Q16): C

0625/41/M/J/25 Q4 — Room heater — convection, current and fuse [9 marks]

4(a)(i) [1] convection
4(a)(ii) [3] warm air rises OR less dense air rises warm air is less dense (than cool air) ORA any one from: • cold air replaces warm air • cold air falls and the process repeats owtte • there is a convection current owtte
4(b)(i) [2] P = IV OR (I =) ■ P V 2.0 kW = 2000 W OR ■ 2000 230 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 11 of 16
4(b)(ii) [3] 10 (A) AND any one from: • smaller fuse melts in normal use (of the heater) owtte • smaller fuse stops the heater working (at all) • larger fuse allows too much current (without melting) • larger fuse may not melt before the circuit is damaged • fuse (rating) must be higher than the (normal) current • the fuse will melt if current goes too high owtte any one from: • 10 (A) • 13 (A) AND fuse (rating) must be higher than (normal) current • 13 (A) AND 3 A / 5 A fuse melts in normal use owtte

0625/42/M/J/25 Q4 — Cooling containers — radiation, conduction and evaporation [8 marks]

4(a)(i) [3] container A or container with (dull) black (surface) AND (dull) black (surfaces) are better emitters of infrared / radiation (than shiny white surfaces) container A or container with (dull) black (surface) AND (dull) black (surfaces) are better emitters (than shiny white surfaces)
4(a)(ii) [1] rate of transfer of energy from the container is equal to rate of transfer of energy to the container
4(a)(iii) [1] convection
4(b) [3] any three from: • metal is a good / better (thermal) conductor OR wood is a poor conductor OR wood is a (thermal) insulator • metal contains free / delocalised electrons OR wood does not contain free / delocalised electrons • electrons carry (thermal) energy through metal OR electrons collide with (distant) ions • wood only transfers (thermal) energy by lattice vibrations 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 13 of 19

Answers — 3 WAVES

3.1 General properties of waves

0625/21/M/J/25 (Q17): B
0625/21/O/N/25 (Q18): C
0625/22/F/M/25 (Q19): D
0625/22/M/J/25 (Q17): C
0625/22/M/J/25 (Q18): B
0625/22/M/J/25 (Q19): D
0625/22/O/N/25 (Q17): D
0625/22/O/N/25 (Q18): B
0625/23/M/J/25 (Q17): D
0625/23/M/J/25 (Q18): C
0625/23/O/N/25 (Q18): C

0625/42/F/M/25 Q7 — Diffraction and wave speed [12 marks]

7(a)(i) [1] diffraction
7(a)(ii) [3] 4.8 ■ 10–7 m OR 480 nm One wavelength marked on Fig. 6.1 OR 1.2 seen
7(a)(iii) [3] at least two curved wavefronts three semi-circular wavefronts centred on (centre of) gap wavelength is unchanged 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED February/March 2025 © Cambridge University Press & Assessment 2025 Page 11 of 14
7(b)(i) [2] v = f■ (■ =) 330 / 380 OR (■ =) 0.87 (m)
7(b)(ii) [3] 7.6 s v = s / t OR (t =) s / v OR (t =) 2500 / 330

0625/42/O/N/25 Q6 — Water waves, refraction and total internal reflection [8 marks]

6(a)(i) [3] wavefronts joined at the boundary angle with the surface in region B smaller than angle with surface in region A AND slanting in correct direction refracted wavefronts all parallel slanting in the correct direction AND wavelength less than incident waves
6(a)(ii) [1] refraction
6(b)(i) [2] speed of light in first region is less than speed of light in second region owtte angle of incidence must be greater than the critical angle
6(b)(ii) [2] any two from: • high rates of data (transmission) • carry large amounts of data / information • secure • little data / signal loss • glass is transparent to visible light and (some) infrared 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 17 of 20

Answers — 3 WAVES

3.2 Light

0625/21/M/J/25 (Q18): D
0625/21/M/J/25 (Q19): B
0625/21/M/J/25 (Q20): B
0625/21/M/J/25 (Q21): B
0625/21/O/N/25 (Q17): B
0625/21/O/N/25 (Q19): B
0625/21/O/N/25 (Q20): B
0625/21/O/N/25 (Q21): D
0625/22/F/M/25 (Q21): A
0625/22/F/M/25 (Q22): D
0625/22/F/M/25 (Q23): A
0625/22/M/J/25 (Q20): D
0625/22/M/J/25 (Q21): A
0625/22/M/J/25 (Q22): D
0625/22/O/N/25 (Q19): C
0625/22/O/N/25 (Q20): C
0625/22/O/N/25 (Q21): A
0625/23/M/J/25 (Q19): B
0625/23/M/J/25 (Q20): D
0625/23/O/N/25 (Q19): A
0625/23/O/N/25 (Q20): A
0625/23/O/N/25 (Q21): C

0625/41/M/J/25 Q5 — Refraction in a soap film — refractive index and frequency [11 marks]

5(a) [1] (refractive index is) the ratio of the speed of light in two different regions owtte OR (refractive index =) ( ) speed of light in air speed of light in soap film
5(b)(i) [2] n = sin sin i r OR (r =) sin–1 sini n ■ ■ ■ ■ ■ ■ OR = sin60 1.28 sinr i = 60 (°)
5(b)(ii) [2] normal drawn (at the point incident ray meets film) refracted ray drawn (refraction towards normal in film) and angle of refraction labelled braille: angle identified but not labelled
5(c)(i) [1] (light of) a single frequency 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 12 of 16
5(c)(ii) [5] 4.4 ■ 1014 Hz (speed of light / e-m waves is approximately) 3.0 ■ 108 m / s (in air) v = f■ OR (f =) – ■ ■ ■ 8 9 3.0 10 680 1 0

0625/41/O/N/25 Q6 — Optical fibre — refractive index and critical angle [11 marks]

6(a) [2] Any two from: • high rates (of data transmission) / faster (data transmission) • carry large amounts (of data / information) • secure • Little data / signal loss • glass is transparent to (some) infrared
6(b)(i) [5] (c =) 46° N = 1 / sin c, 1.4 = 1 / sin c OR (c =) sin–1 (1 / 1.4) Equation in this form only (c =) sin–1 (1 / n) OR (c =) sin–1 (1 / 1.4) OR (c =) 46
6(b)(ii) [1] angle of incidence (of light) at which the angle of refraction is (exactly) 90° 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 15 of 18
6(b)(iii) [3] angle of incidence marked AND TIR of ray inside glass AND light ray leaves end of fibre after two or three reflections angle of incidence marked OR TIR drawn until ray leaves end of fibre

0625/42/F/M/25 Q6 — Converging lens — ray diagram and image characteristics [9 marks]

6(a)(i) [1] Horizontal axis labelled P
6(a)(ii) [1] X on horizontal axis 3.0 cm to the left of centre of L OR X on horizontal axis 3.0 cm to the right of centre of L
6(a)(iii) [2] Any two from: • Straight line from a point on O, passing through centre of L (and beyond) • Horizontal line from same point on O to L, refracted through F (on RH side of L) • Straight line from same point on O through F (on LH side of L) to L, refracted parallel to principal axis both rays

extended until they intersect 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED February/Ma
Cambridge University Press & Assessment 2025 Page 10 of 14
6(a)(iv) [3] diminished enlarged inverted real same size upright virtual
6(b) [2] (image is now) virtual (image is now) upright

0625/42/M/J/25 Q6 — Converging lens and correcting long-sightedness [9 marks]

6(a)(i) [1] two points labelled F on principal axis, both 3.0 cm from the centre of the lens
6(a)(ii) [5] method 1 two correct rays from: • ray from top of I through the centre of the lens • ray from top of I parallel to principal axis to the left of the lens, to centre of lens and then through RH focus • ray from top of I through LH focus to lens, to centre of lens and then parallel to principal axis on the right of the lens OR method 2 two correct rays from: • ray from top of I through the centre of the lens • ray from top of I through RH focus of lens and extended back, from centre of lens, parallel to principal axis on the left of the lens • ray from top of I parallel to principal axis to the left of the lens, and extended back, from the centre of the lens, through the LH focus object, labelled O, at the intersection of rays to the RH of the lens for method 1 object, labelled O, at the intersection of rays to the LH of the lens for method 2
6(a)(iii) [1] real OR inverted
6(b)(i) [1] rays meeting behind the retina OR would meet behind retina if extended
6(b)(ii) [1] any converging lens 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 16 of 19

0625/43/M/J/25 Q5 — Light in a swimming pool — refraction and total internal reflection [8 marks]

5(a) [1] (refractive index is) the ratio of the speed of light in two different mediums OR (refractive index =) speed of light in air speed of light in water OR (refractive index =) sine of angle of incidence sine of angle of refraction
5(b) [2] P continues vertical Q refracted away from normal in the correct direction
5(c) [3] 1.3 (n =) 1 ÷ sin 49 OR (n =) 1 ÷ sin c
5(d) [2] any two from: • (ray travelling from) dense to less dense medium OR water is more dense than air • critical angle = 49° • angle of incidence exceeds critical angle / 49°

Answers — 3 WAVES

3.3 Electromagnetic spectrum

0625/21/M/J/25 (Q22): D
0625/21/O/N/25 (Q22): D
0625/22/M/J/25 (Q23): C
0625/22/O/N/25 (Q22): B
0625/23/M/J/25 (Q21): C
0625/23/O/N/25 (Q22): D

0625/43/O/N/25 Q5 — Electromagnetic spectrum, waves and seismic P-waves [10 marks]

5(a)(i) [2] all four correct = 2 marks; any two correct = 1 mark security marking gamma rays Bluetooth ultraviolet optical fibres infrared detection of cancer radio waves

5(a)(ii) [1] any one from: • only require a short aerial / antenna • can pass through the atmosphere / ionosphere to satellites • can transmit large amounts of data at high speeds / high rate of data transmission

5(b) [2] any two from: transverse: • oscillations / vibrations are at right angles to direction of propagation • medium not required / can travel through a vacuum (for e-m wave) • have crests / peaks and troughs longitudinal: • oscillations / vibrations are parallel to direction of propagation • medium is required • have rarefactions and compressions

5(c)(i) [1] longitudinal

5(c)(ii) [2] refraction any one from: • change in speed of the wave • change in density of the medium

5(d) [2] increase in carbon dioxide / greenhouse gases (in the atmosphere) OR increase in methane / water vapour (in the atmosphere) any one from: prevents / reduces (re-emitted) radiation escaping (into space) OR radiation re-emitted / reflected back to surface (by atmosphere / greenhouse gases) OR (amount of) radiation absorbed is greater than (the amount of) radiation leaving the atmosphere

Answers — 3 WAVES

3.4 Sound

0625/21/M/J/25 (Q23): D
0625/21/O/N/25 (Q23): B
0625/22/F/M/25 (Q20): A
0625/22/M/J/25 (Q24): C
0625/22/M/J/25 (Q25): D
0625/22/O/N/25 (Q23): C
0625/23/M/J/25 (Q22): C
0625/23/M/J/25 (Q23): A
0625/23/O/N/25 (Q23): D

0625/41/O/N/25 Q5 — Dolphin sounds — speed, frequency, loudness and pitch [9 marks]

5(a) [1] (speed of sound in air =) 330 m / s ■ value ■ 350 m / s AND (speed of sound in water) is faster
5(b) [3] Yes AND (normal) human hearing range is 20 Hz to 20 kHz (and all dolphin sounds lie in this range) Yes AND all the dolphin sounds are in the range of human hearing 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 14 of 18
5(c) [2] Frequency kHz amplitude loudness pitch 14 Large loud high 8 small soft / quiet low one mark for each column correct
5(d) [1] Particles
5(d) [2] high pressure AND low pressure Parallel

0625/42/M/J/25 Q5 — Speed of sound and Bluetooth microwaves [19 marks]

5(a) [12] echo method (outside): 1 method of producing short, loud sound (e.g. clap / shout / cry out / bang 2 pieces of wood together) 2 measuring tape or trundle wheel, stopwatch and wall 3 (person) starts stopwatch when clap etc heard and stops it when echo heard 4 speed = 2 ■ measured distance ÷ time OR direct method outside: 1 method of producing short, loud sound (e.g. fire gun) 2 measuring tape or trundle wheel, stopwatch 3 (student at one end) starts stopwatch when smoke seen from gun and stops it when sound heard 4 speed = distance ÷ time OR direct method (using digital): 1 method of producing short sound e.g. clap or hammer striking block 2 measuring tape or metre ruler, digital timer and microphones 3 digital timer starts when sound reaches first microphone and stops when sound reaches second microphone. Time difference is recorded on digital timer owtte 4 speed = distance ÷ time for direct method 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 14 of 19
5(b)(i) [1] radio waves
5(b)(ii) [1] 3(.0) ■ 108 m / s
5(b)(iii) [5] 0.12 m v = f■ OR (■ =) v ÷ f OR (■ =) {3(.0) ■ 108} ÷ {2.48 ■ 109} (■ =) {3(.0) ■ 108} ÷ {2.48 ■ 109} OR (■ =) 1.2 ■ 10N 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 15 of 19

0625/43/M/J/25 Q6 — Ultrasound — longitudinal waves and locating objects [6 marks]

6(a) [1] sound with a frequency higher than 20 kHz
6(b) [1] vibrations (of the wave / particles) are parallel to the direction of propagation
6(c)(i) [3] (pulse of) ultrasound / sound / wave (sent into water) reflects from object time to travel to object and back measured depth = speed ■ time 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 11 of 13
6(c)(ii) [1] any one from: • non-destructive testing of materials • medical scanning (of soft tissue)

Answers — 4 ELECTRICITY AND MAGNETISM

4.1 Simple phenomena of magnetism

0625/21/M/J/25 (Q24): A

0625/21/O/N/25 (Q24): D

0625/22/O/N/25 (Q24): A

0625/23/M/J/25 (Q24): B

0625/41/M/J/25 Q6 — Magnetic window cleaner — magnets and field lines [4 marks]

6(a) [1] any one from: • magnetic materials are attracted to magnets • non-magnetic materials are not attracted to magnets • magnetic materials experience a force in magnetic fields • non-magnetic materials don’t experience a force in magnetic fields

6(b) [1] any one from: • steel as it stays magnetised • steel makes a permanent magnet

6(c) [1] N and S poles correctly labelled braille: north to south owtte

6(d) [1] field lines close(r) indicates strong(er) (magnetic) field owtte ORA

Answers — 4 ELECTRICITY AND MAGNETISM

4.2 Electrical quantities

0625/21/M/J/25 (Q25): A
0625/21/M/J/25 (Q26): D
0625/21/O/N/25 (Q25): D
0625/21/O/N/25 (Q26): A
0625/21/O/N/25 (Q27): C
0625/22/F/M/25 (Q24): B
0625/22/F/M/25 (Q25): A
0625/22/F/M/25 (Q26): A
0625/22/F/M/25 (Q27): D
0625/22/M/J/25 (Q26): A
0625/22/M/J/25 (Q27): D
0625/22/M/J/25 (Q28): C
0625/22/O/N/25 (Q25): B
0625/22/O/N/25 (Q26): C
0625/23/M/J/25 (Q25): D
0625/23/M/J/25 (Q26): D
0625/23/M/J/25 (Q27): B
0625/23/O/N/25 (Q25): A
0625/23/O/N/25 (Q26): C
0625/23/O/N/25 (Q27): B
0625/23/O/N/25 (Q28): B

0625/42/F/M/25 Q3 — Electric heater — radiation, earthing and electrical power [15 marks]

3(a) [1] infrared
3(b) [3] shiny surface / it is a good reflector of radiation Any one from: • it is a good reflector • it reflects radiation
3(c) [1] Any one from: • prevents (electric) shock (if live wire touches the metal casing) owtte • if live wire touches the metal casing the current goes to earth
3(d)(i) [3] 2.6 A R = V / I OR (I=) V/R OR (I=) 230 / 89
3(d)(ii) [2] P = IV (I =) 5.2 (A) OR (P =) 2 ■ power of one element OR
3(d)(iii) [5] 68 000 J OR 68 kJ E = Pt OR (E =) Pt OR (E =) 1200 ■ 60 efficiency = useful energy out / total energy (in) OR 95 100 E ■ ■ (power output of heater =) 95% ■ 1200 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED February/March 2025 © Cambridge University Press & Assessment 2025 Page 8 of 14

0625/42/M/J/25 Q7 — Electric field, charging by friction and electrical energy [13 marks]

7(a) [1] (region) where a(n electric) charge experiences a force OR (region) where a force acts on a(n electric) charge
7(b)(i) [2] any two from: • friction (between cloth and rod causes electrons to gain energy) • electrons move • (electrons move) from cloth / to plastic (making plastic negative and cloth positive)
7(b)(ii) [3] moves away (from the charged plastic rod) moves / experiences a force / repels
7(c)(i) [1] energy transferred in one hour at a rate of transfer of 1 kW
7(c)(ii) [6] 1 5.5 (kW h) P = (∆)E ÷ t OR (∆E =) Pt OR (∆E =) 0.025 ■ 220 OR 25 ■ 220 2 0.11 ■ A P = V I OR (I =) P ÷ V OR (I =) 25 ÷ 230 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 17 of 19

0625/43/M/J/25 Q8 — Electric field and charge calculations [11 marks]

8(a)(i) [1] (region) where (an electric) charge experiences a force OR (region) where a force acts on a (an electric) charge
8(a)(ii) [2] four radial straight field lines starting at charge at least one arrow on a field line towards the charge 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 12 of 13
8(b) [5] 1600 C V = W / Q OR (Q =) W / V (Q =) 4.5 ■ 10N ÷ 2.9 ■ 108 OR 1.6 ■ 10N
8(c) [3] 240 C (Q =) It OR 2(.0) ■ 120

Answers — 4 ELECTRICITY AND MAGNETISM

4.3 Electric circuits

0625/21/M/J/25 (Q27): C
0625/21/M/J/25 (Q28): C
0625/21/O/N/25 (Q28): D
0625/21/O/N/25 (Q29): C
0625/22/F/M/25 (Q28): C
0625/22/F/M/25 (Q29): D
0625/22/M/J/25 (Q29): B
0625/22/M/J/25 (Q30): A
0625/22/M/J/25 (Q31): D
0625/22/O/N/25 (Q27): D
0625/22/O/N/25 (Q28): B
0625/23/M/J/25 (Q28): C
0625/23/O/N/25 (Q29): D
0625/23/O/N/25 (Q30): D
0625/23/O/N/25 (Q31): C

0625/41/M/J/25 Q7 — Current–voltage graph and a parallel heater circuit [13 marks]

7(a)(i) [2] diode there is only a current (in diode) when the voltage is increased in one direction owtte OR only a current in one direction
7(a)(ii) [1] braille: symbol A identified 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 13 of 16
7(b)(i) [3] 59 ■ V = IR OR (R =) ■ V I OR (R =) ■ 230 3.9
7(b)(ii) [5] 270 000 J OR 2.7 ■ 105 J (E =) IVt (t =) ■ 5 60 seen
7(b)(iii) [2] 7.8 A any one from: • identical heaters (each with p.d. of 230 V) so 3.9 A in each branch • I = I1 + I2 • 1 2 1 1 1 = + R R R in any form

0625/41/O/N/25 Q7 — Circuit with resistors — current and resistance [11 marks]

7(a) [3] 38 ■ (R =) V ÷ I OR (R =) 3 ÷ 0.08(0)
7(b)(i) [6] Current in lower branch is 0.16(0) A ammeter reading is sum of currents in each branch (ammeter reading =) 0.24 A OR resistance in top branch = twice value in 7(a) formula for combined resistance of resistors in parallel Correct current calculated from candidate's R1 in 7(a)
7(b)(ii) [2] potential difference (p.d.)(across R3) is larger (than p.d. across R1) more work done (passing charge through R3) 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 16 of 18

0625/42/F/M/25 Q5 — Light-dependent resistor — resistance and a light circuit [7 marks]

5(a) [2] y-axis labelled resistance AND x-axis labelled light intensity Straight line / smooth curve with negative gradient
5(b)(i) [1] Correct symbol drawn to complete circuit. 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED February/March 2025 © Cambridge University Press & Assessment 2025 Page 9 of 14
5(b)(ii) [4] in the dark, VLDR is a bigger proportion of e.m.f OR when RLDR is high VLDR is a bigger proportion of e.m.f. OR when VLDR is high VLDR is a bigger proportion of e.m.f. In the dark, VLDR is high OR when RLDR high, VLDR is high Any one from: • emf shared (between fixed resistor and LDR) OR emf is constant • VLDR = VLAMP OR p.d. is the same across components in parallel

0625/42/O/N/25 Q7 — Parallel lamps — current, energy and combined resistance [13 marks]

7(a) [2] correct symbols for a.c supply and lamps components all connected in parallel
7(b)(i) [3] 0.29 A (I =)V / R OR (I =)230 / 800
7(b)(ii) [5] 0.33 (kWh) (E =)VIt OR (E =) {230 ■ 0.29 ■ 5(.0)} / 1000 conversion from W to kW seen i.e. division by 1000
7(b)(iii) [3] 480 ■ 1 / R = 1 / R1 + 1 / R2 OR 1 / R = 1 / 800 + 1 / 1200 OR (R =) R1R2 / {R1 + R2} OR (R =) {800 ■ 1200} / {800 + 1200} 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 18 of 20

0625/43/M/J/25 Q7 — Circuit calculations — current, p.d. and resistance [10 marks]

7(a) [1] 0.5(0) A
7(b) [3] 3(.0) V (V =) IR OR 0.5(0) ■ 6(.0)
7(c) [1] 12 – 3(.0)
7(d) [5] 180 ■ I2 = 9.0 ÷ 20 OR I2 = 0.45 (A) OR I1 = 0.05 (A) (R =) 9 ÷ 0.05

0625/43/O/N/25 Q6 — LDR and LED circuits — current and p.d. [12 marks]

6(a)(i) [5] 0.008(0) A OR 8(.0) mA OR 8(.0) ■ 10–3 A (p.d. across 400 ■ resistor =) 5 – 1.8 OR 3.2 (V)
(I =) V / R OR 3.2 / 400
6(a)(ii) [3] increases resistance of LDR increases ratio of potential differences (across each component) equals ratio of the two resistances OR R1 / R2 = V1 / V2 OR LDR has greater share of the total / supply voltage OR potential difference of the source is divided between the resistor (R) and LDR
6(b)(i) [3] 56 mA OR 5.6 ■ 10–2 A 44 (mA) OR 12 (mA)
6(b)(ii) [1] 0 (A) 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 16 of 20

Answers — 4 ELECTRICITY AND MAGNETISM

4.4 Electrical safety

0625/22/O/N/25 (Q29): A
0625/23/M/J/25 (Q29): B

Answers — 4 ELECTRICITY AND MAGNETISM

4.5 Electromagnetic effects

0625/21/M/J/25 (Q29): C
0625/21/M/J/25 (Q30): B
0625/21/M/J/25 (Q31): A
0625/21/M/J/25 (Q32): D
0625/21/O/N/25 (Q30): D
0625/21/O/N/25 (Q31): D
0625/21/O/N/25 (Q32): C
0625/22/F/M/25 (Q30): C
0625/22/F/M/25 (Q31): B
0625/22/F/M/25 (Q32): C
0625/22/F/M/25 (Q33): C
0625/22/M/J/25 (Q32): B
0625/22/M/J/25 (Q33): C
0625/22/O/N/25 (Q30): B
0625/22/O/N/25 (Q31): C
0625/22/O/N/25 (Q32): D
0625/23/M/J/25 (Q30): A
0625/23/M/J/25 (Q31): A
0625/23/M/J/25 (Q32): D
0625/23/O/N/25 (Q24): C
0625/23/O/N/25 (Q32): D

0625/41/M/J/25 Q8 — a.c. generator and high-voltage transmission [12 marks]

8(a)(i) [2] (A is carbon) brushes (B is) slip rings
8(a)(ii) [1] strengthens the magnetic field (of the magnet)
8(b) [3] graph is sinusoidal with positive and negative e.m.f. graph shows minimum of one cycle completed in 0.5 s e.m.f. is a maximum at 0.00 s, curves to a minimum at 0.25 s and curves to maximum at 0.50 s OR e.m.f. is a minimum at 0.00 s and curves to a maximum at 0.25 s and curves to a minimum at 0.50 s braille: e.m.f. starts at time zero at either maximum or minimum value 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 14 of 16
8(c)(i) [3] either: less power loss (for same power transmission) AND (because) P = I2R OR low current (for same power transmission) AND (allows) thinner / cheaper cables any one from: • less power loss (for same power transmission) • P = I2R • low current (for same power transmission) • thinner / cheaper cables
8(c)(ii) [3] 5400 p p s s = V N V N OR (Ns =) p s p N V V OR (Ns =) 450 × 300 000 25 000 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 15 of 16

0625/41/O/N/25 Q8 — Solenoid and an electric bell — magnetic field [10 marks]

8(a)(i) [4] field lines parallel inside solenoid Minimum of 2 magnetic field lines, curved towards end of solenoid, around outside of solenoid Correct symmetrical pattern above and below solenoid AND no field lines crossing AND At least 2 complete lines At least one arrow on a field line outside the solenoid which is away from left end or towards right end of solenoid
8(a)(ii) [1] B marked at a place where field lines are close together in Fig. 8.1 AND (magnetic) field lines are close(r) together
8(b)(i) [1]
8(b)(ii) [2] Any one from: • solenoid becomes an (electro)magnet • there is a magnetic field (around solenoid) Any one from: • attraction between solenoid and the (soft) iron (arm) • (soft) iron becomes an (induced) magnet
8(b)(iii) [2] Any one from: • contacts are broken • circuit is broken (when striker hits bell) • there is no current (in the solenoid) Any one from: • solenoid stops being a magnet • soft iron loses magnetism • soft iron stops being attracted to solenoid • Springy metal moves the soft iron back owtte 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press &

0625/42/F/M/25 Q8 — Motor effect and the electric motor [8 marks]

8(a) [3] (direction of force) down(wards) magnetic field direction, current direction and force are mutually perpendicular magnetic field is from N to S / left to right AND current flows from positive to negative / anticlockwise / into paper
8(b) [1] (direction of force) reverses / changes by 180° 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED February/March 2025 © Cambridge University Press & Assessment 2025 Page 12 of 14
8(c) [4] (J carbon brushes) Any one from: • connect cell / circuit to coil / wire / split ring(s) / commutator • maintains (continuous) connection • prevent wires from tangling (as motor rotates) (K coil) Any one from: • rotates / turns • conducts / has a current in it (L axle) Any one from: • allows the coil to rotate / turn • allows motor to turn / spin (M split ring commutator) Any one from: • keeps motor turning in the same direction owtte • reverses the connections to the coil (every half-turn) owtte • prevents wires from tangling (as the motor rotates)

0625/42/M/J/25 Q8 — a.c. generator and a resistor circuit [10 marks]

8(a)(i) [2] P and Q: slip rings X and Y: brushes
8(a)(ii) [1] coil cuts magnetic field
8(a)(iii) [1] any one from: • increase strength of magnetic field • increase speed of rotation of coil • increase number of turns (of coil)
8(b)(i) [2] 1.2 A (total) resistance of two (identical) resistors in series is added / doubled
8(b)(ii) [1] 6(.0) A
8(b)(iii) [3] 2.1 ■ V = I R OR (R =) V ÷ I OR (R =) 5(.0) ÷ 2.4 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 18 of 19

0625/42/O/N/25 Q8 — Magnetic field and electromagnetic induction [5 marks]

8(a) [1] region in which a (magnetic) pole experiences a force
8(b)(i) [1] friction between trolley and track OR the (induced) current produces a magnetic field which opposes the (magnetic) field that causes the (induced) current
8(b)(ii) [3] any three from: • direction will (still) be left OR there is no change of direction • induced current / emf / voltage does not change direction • smaller deflection • smaller (rate of) change of magnetic field • smaller current / emf / voltage induced (in coil) 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 19 of 20

0625/43/M/J/25 Q9 — Transformer — alternating current and high-voltage transmission [11 marks]

9(a)(i) [3] any three from: • (current in the primary coil generates a) changing magnetic field (in primary coil) • (iron) core transfers the magnetic field (to the secondary coil) • secondary coil cuts the magnetic field / secondary coil is in (changing) magnetic field • an e.m.f. is induced (in the secondary coil) • (induced) current changes direction because the magnetic field changes direction
9(a)(ii) [3] 240 V Vs ÷ Vp = Ns ÷ Np OR (Vs =) 12 ■ 20(÷ 1)
9(b) [3] 0.16 A P = I2R OR I2 = P ÷ R OR I2 = 1.25 ■ 10–3 ÷ 0.05(0) OR I2 = 0.025
9(c) [2] any two from: • less power / heating / energy losses • thinner / cheaper cables • pylons further apart / fewer pylons • transfer energy over long(er) distance 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 13 of 13

0625/43/O/N/25 Q7 — a.c. generator — slip rings and e.m.f. [10 marks]

7(a)(i) [1] provide a continuous connection (between the coil and the lamp) OR prevent the contacts reversing every half turn / 180° OR prevent the wires (to the lamp) twisting / tangling
7(a)(ii) [3] any three from: • coil is rotated / moved in magnetic field • coil cuts (magnetic) field OR changing (magnetic) field through coil • e.m.f. / voltage induced • coil forms part of a complete circuit (through the lamp) so there is a current • as one side of the coil starts to move in the opposite direction, current is induced in the opposite direction OR direction of current in coil is reversed every half turn / 180°
7(a)(iii) [2] any two from: • increase strength of magnetic field • increase speed of rotation (of the coil) • increase number of turns (of coil)
7(b)(i) [1] 2.5 (revolutions) 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 17 of 20
7(b)(ii) [3] any 1 correct = 1 mark any 2 or 3 correct = 2 marks all 4 correct = 3 marks Time Label from graph coil position A B C D N S N S N S N S 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED

Answers — 5 NUCLEAR PHYSICS

5.1 The nuclear model of the atom

0625/21/M/J/25 (Q33): A
0625/21/M/J/25 (Q34): B
0625/21/O/N/25 (Q33): B
0625/22/M/J/25 (Q34): B
0625/22/O/N/25 (Q33): B
0625/22/O/N/25 (Q34): D
0625/23/M/J/25 (Q34): C

0625/41/O/N/25 Q9 — Helium-4 — atomic structure, fusion and stars [9 marks]

9(a) [3] Any three from: • nucleus containing protons and neutrons • protons (in nucleus) • 2 neutrons (in nucleus) • 2 electrons outside the nucleus • Protons are positive (charges) OR electrons are negative (charges)

9(b)(i) [2] hydrogen (hydrogen) nuclei join (together)

9(b)(ii) [2] Any two from: • infrared • visible • ultraviolet

9(c) [2] • (the star) expands Any one from: • (smaller star) forms a red giant • (more massive star) forms a red supergiant 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 18 of 18

Answers — 5 NUCLEAR PHYSICS

5.2 Radioactivity

0625/21/M/J/25 (Q35): C
0625/21/M/J/25 (Q36): A
0625/21/M/J/25 (Q37): D
0625/21/O/N/25 (Q34): B
0625/21/O/N/25 (Q35): C
0625/21/O/N/25 (Q36): C
0625/21/O/N/25 (Q37): A
0625/22/F/M/25 (Q34): C
0625/22/F/M/25 (Q35): A
0625/22/F/M/25 (Q36): A
0625/22/F/M/25 (Q37): D
0625/22/F/M/25 (Q38): D
0625/22/M/J/25 (Q35): D
0625/22/M/J/25 (Q36): A
0625/22/M/J/25 (Q37): C
0625/22/O/N/25 (Q35): D
0625/22/O/N/25 (Q36): A
0625/22/O/N/25 (Q37): B
0625/23/M/J/25 (Q33): D
0625/23/M/J/25 (Q35): C
0625/23/M/J/25 (Q36): A
0625/23/M/J/25 (Q37): C
0625/23/O/N/25 (Q33): A
0625/23/O/N/25 (Q34): B
0625/23/O/N/25 (Q35): B
0625/23/O/N/25 (Q36): B
0625/23/O/N/25 (Q37): D

0625/41/M/J/25 Q9 — Strontium-90 — beta decay and half-life [14 marks]

9(a)(i) [3] (stable isotope) has fewer neutrons AND radioactive isotopes (usually) have an excess of neutrons OR (stable isotope) has fewer neutrons AND radioactive isotopes are too heavy any one from: • (stable isotope) has fewer neutrons • radioactive isotopes (usually) have more neutrons • radioactive isotopes are too heavy

9(a)(ii) [2] 90 39Y 0 –1■

9(a)(iii) [3] ionising radiation is harmful (to humans) OR beta particles are ionising and harmful (to humans) any one from: • radiation is / beta particles are harmful • beta particles ionise • it ionises AND is harmful

9(b)(i) [5] 72 ■ half-life ■ 76 (h) braille: 80 h (mark based on candidate choice of halving) evidence of count rate halved e.g. 48 ■ 2 = 24 evidence on graph or in working that Fig. 9.1 is used to find time for count rate to halve

9(b)(ii) [1] any one from: • table includes background radiation owtte • graph does not have background count rate owtte • graph has corrected count rate 0625/41 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 16 of 16

0625/42/F/M/25 Q9 — Strontium-90 — isotopes, beta decay and half-life [8 marks]

9(a)(i) [1] (isotopes are) forms of an element with the same number of protons but a different number of neutrons (in the nucleus) OR (isotopes are) the same element with a different number of neutrons 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED February/March 2025 © Cambridge University Press & Assessment 2025 Page 13 of 14

9(a)(ii) [2] particle number in each atom of strontium location electron 38 outside nucleus neutron proton 38 Inside nucleus particle number in each atom of strontium location 38 outside nucleus neutron 52 Inside nucleus 38 Inside nucleus

9(b)(i) [3] (there is a) different count rate with different thicknesses of metal OR number of ■-particles detected varies with thickness (beta (■) particles) can penetrate thin / 0.75mm metal OR (beta (■) particles) are stopped by thick metal
9(b)(ii) [2] (approximate) percentage of source remaining (after 15 years) stated AND in range 63–7 (approximate) percentage of sample lost (after 15 years) stated AND in range 30–37% Any one fron count rate / activity (too) low (to detect differences in thickness) owtte • detector needs high activity (to detect differences in thickness) owtte • count rate / activity too close to background owtte • less difference in activity for different thicknesses owtte 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED February/March 2025 © Cambridge University Press & Assessment 2025 Pa

0625/42/M/J/25 Q9 — Radium and technetium — medical radioactive isotopes [7 marks]

9(a)(i) [1] radiation (always) present in the environment OR radiation from natural sources
9(a)(ii) [1] any one from: • radon gas • rocks OR buildings • food OR drink • cosmic rays
9(b)(i) [2] Rn 86 219 ■ 2 4
9(b)(ii) [1] alpha (particles) would be absorbed / stopped by the skin owtte
9(c)(i) [1] gamma / radiation needs to pass out of the body (to detector)
9(c)(ii) [1] after a few days / some time, little radiation is emitted owtte 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 19 of 19

0625/42/O/N/25 Q9 — Radiation in a magnetic field and choosing a radioactive source [9 marks]

9(a)(i) [1] curve bending upwards while in magnetic field and labelled ■
9(a)(ii) [2] curve bending downwards while in magnetic field and labelled ■ smaller radius of curvature for β than for α
9(a)(iii) [1] line passing straight through the magnetic field and labelled ■
9(b)(i) [1] time taken for half the nuclei of a sample to decay
9(b)(ii) [4] R half-life not too short OR half-life is suitable so that source does not need to be replaced often so count rate changes with thickness γ- rays would not be stopped at all AND α-particles would be absorbed by any thickness of aluminium 0625/42 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 20 of 20

0625/43/M/J/25 Q11 — Half-life, beta decay and alpha ionisation [4 marks]

11(a) [1] time taken for half the nuclei (in any sample) to decay
11(b)(i) [0]
11(b)(ii) [1] neutron changes to a proton (plus an electron)
11(c) [2] (■ particles have) greater kinetic energy (than ■ particles) (■ particles have) greater charge (than ■ particles)

0625/43/O/N/25 Q8 — Alpha scattering, radiation deflection and half-life [8 marks]

8(a)(i) [1] most of the atom is empty space OR nucleus is (very) much smaller than the atom
8(a)(ii) [1] nucleus must have a positive charge OR (the positive) charge is concentrated at the centre of the atom OR most of the mass (of the atom) is concentrated in the nucleus
8(b)(i) [2] P alpha particles have greater mass so are deflected less (than the beta particles)
8(b)(ii) [1] upwards AND alpha particles are positively charged so attracted towards negatively charged plate OR direction of field is direction of force on positive charge
8(c) [3] 1.9 ■ 10–2 g OR 1.9 ■ 10–5 kg 3 half-lives OR 2.4 ■ 10–3 ■ 23 OR 1 / 8 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED October/November 2025 © Cambridge University Press & Assessment 2025 Page 19 of 20

Answers — 6 SPACE PHYSICS

6.1 The Earth and the Solar System

0625/21/M/J/25 (Q38): B
0625/21/O/N/25 (Q38): C
0625/21/O/N/25 (Q39): D
0625/22/M/J/25 (Q38): C
0625/23/M/J/25 (Q38): A
0625/23/M/J/25 (Q40): A
0625/23/O/N/25 (Q38): D

0625/41/M/J/25 Q10 — Jupiter and Earth — gravitational field and orbital speed [9 marks]

10(a) [2] Jupiter is gaseous Earth is rocky
10(b)(i) [3] (gravitational) force per unit mass OR (g =) weight mass in this form (gravitational) force on a mass OR W = mg
10(b)(ii) [1] mass
10(c) [3] (orbital speed of) Jupiter is slower ORA Jupiter is further from the Sun ORA OR orbital speeds of planets decrease as distance from the Sun increases ORA gravitational field (strength) of Sun decreases with distance (from Sun) ORA

0625/41/O/N/25 Q10 — Comet orbit and light-years [7 marks]

10(a) [1] B AND Sun is not at the centre of the orbit OR comets have an elliptical orbit
10(b) [2] Comet travels faster when it is closer to the Sun ORA GPE decreases AND KE increases as comet gets closer to the Sun ORA
10(c)(i) [1] distance travelled (in the vacuum of space) by light in one year.
10(c)(ii) [3] 6.3 ■ 1010 (m) OR 63 000 000 000 (m) (1 light-year =) 9.5 ■ 1015 (m) OR (1 light-year =) 9.5 ■ 1012 (km) OR SEEN (distance =) 6.6 ■ 10–6 ■ 9.5 ■ 10N

0625/42/F/M/25 Q10 — Earth's orbit — days, seasons and orbital speed [10 marks]

10(a) [2] (number of days =) 91 (statement F to G) is ¼ of (a complete) orbit OR a whole year is 365 days
10(b)(i) [1] F
10(b)(ii) [1] H
10(c) [5] 1.5 ■ 1011 m v = 2■r/T OR (r =) vT/(2■) OR ( ) 4 3.0 1 0 365 24 60 60 / 2 r ■ = ■ ■ ■ ■ ■ (T =) 365 ■ 24 ■ 60 ■ 60 OR (T =) 31 536 000 OR (r =) 3.0 ■ 104 ■ T / (2■) OR correct rearrangement of and substitution into formula using candidate's value of T
10(d) [1] Any one from: • minor planets • asteroids OR meteoroids • moons (that orbit planets) • comets • natural satellites

0625/42/M/J/25 Q10 — Planets — temperature, gravity and the life cycle of a star [10 marks]

10(a)(i) [2] planet B high(est) (surface) temperature
10(a)(ii) [1] –180 (°C)
10(a)(iii) [2] planet A gravitational field strength is highest so acceleration (due to gravity) is greater OR gravitational field strength is highest so greater (downward) force on the object
10(b)(i) [4] red supergiant (new) heavier elements OR hydrogen any two from: • neutron star • black hole • new stars (with orbiting planets)
10(b)(ii) [1] distance (from Earth) to galaxy

0625/42/O/N/25 Q10 — Solar System formation and the Whirlpool galaxy [14 marks]

10(a) [4] rocky gaseous gravitational force accretion disc
10(b) [5] 3.9 ■ 108 (m) (speed of light =) 3.0 ■ 108 (s =) vt OR (s =) 3.0 ■ 108 ■ 1.3
10(c) [5] 4.8 ■ 105 m / s OR 4.8 ■ 102 km / s (1 light-year =) 9.5 ■ 1015 (m) OR 9.5 ■ 1012 (km) OR 4.8 ■ 10N (v =) H0 d OR (v =) 2.2 ■ 10-18 ■ 2.3 ■ 107 ■ 9.5 ■ 1015

0625/43/O/N/25 Q9 — CMBR and planetary orbits [9 marks]

9(a)(i) [1] (shortly after) Big Bang OR when Universe was formed
9(a)(ii) [2] Increased Universe has expanded OR Universe accelerated outwards
9(b)(i) [5] 120 000 (km / h) OR 1.2 ■ 105 (km / h) (v =) 2■r / T OR {2 ■ ■ ■ 108 (■ 106) } / {0.62 ■ 365 ■ 24} (T =) 0.62 ■ 365 ■ 24 OR (T =) 5431 OR (T =) 226.3 ■ 24 OR 1.2 ■ 10n

9(b)(ii) [1] decreases 0625/43 Cambridge IGCSE – Mark Scheme PUBLISHED October/November University Press & Assessment 2025 Page 20 of 20

0625/43/O/N/25 Q10 — Comet orbit — speed and conservation of energy [6 marks]

10(a) [2] elliptical shape drawn S at a focus of the ellipse by eye

10(b) [1] L furthest away from their S / Sun

10(c) [3] total energy = gravitational (potential) energy + kinetic energy (energy stored as) GPE decreases and (energy stored as) KE increases closer to the Sun when the comet has more kinetic energy it has a greater speed S OR S S L

Answers — 6 SPACE PHYSICS

6.2 Stars and the Universe

0625/21/M/J/25 (Q39): C
0625/21/M/J/25 (Q40): C
0625/21/O/N/25 (Q40): B
0625/22/F/M/25 (Q40): A
0625/22/M/J/25 (Q39): D
0625/22/M/J/25 (Q40): D
0625/22/O/N/25 (Q39): B
0625/22/O/N/25 (Q40): A
0625/23/M/J/25 (Q39): C
0625/23/O/N/25 (Q17): A
0625/23/O/N/25 (Q39): C
0625/23/O/N/25 (Q40): A

0625/43/M/J/25 Q10 — Light-years, the Hubble constant and redshift [11 marks]

10(a) [1] distance travelled (in the vacuum of space) by light in one year

10(b) [3] 1.5 ■ 1011 m 9.5 ■ 1015 (m) OR (c =) 3(.0) ■ 108 (m / s) OR 3(.0) ■ 10N ■ 490

10(c) [5] 2.5 ■ 10–18 per second (Hubble's constant =) gradient OR (H0 =) v ÷ d 2.5 ■ 104(– 0) ÷ {100 ■ 1020 (– 0)}

10(d) [2] electromagnetic radiation / light from (distant) galaxies (observed) increase in wavelength (compared to wavelength measured on the Earth)